

能源电化学

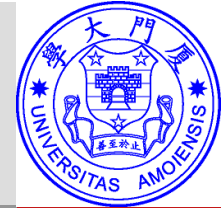
化学化工学院

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锂离子电池的负极材料

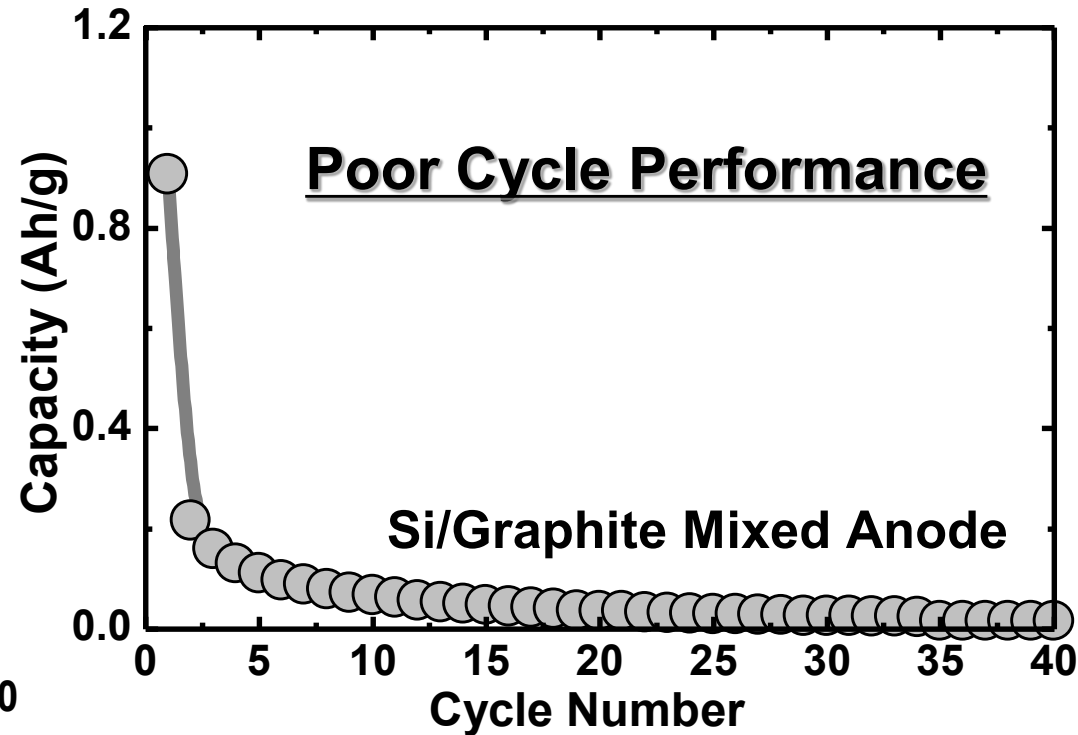
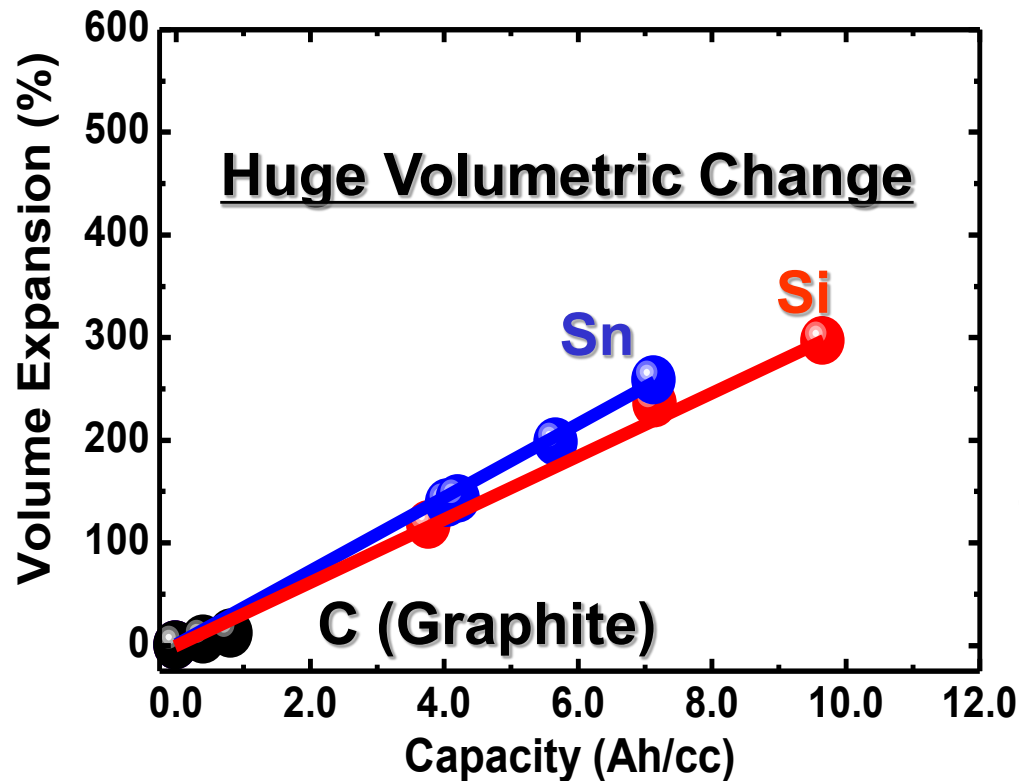


- 体积变化大
- 电子导电性差
- 初期不可逆容量高

如何解决？

- 纳米化
- 碳包覆
- 空隙率调控

Si shows High Capacity (4.2Ah/g) with Low Voltage (0.4V), but ..



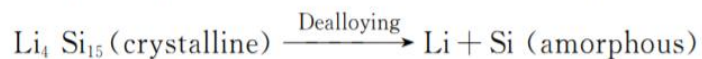
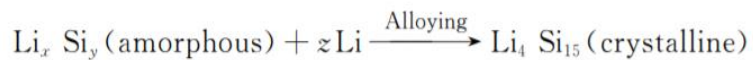
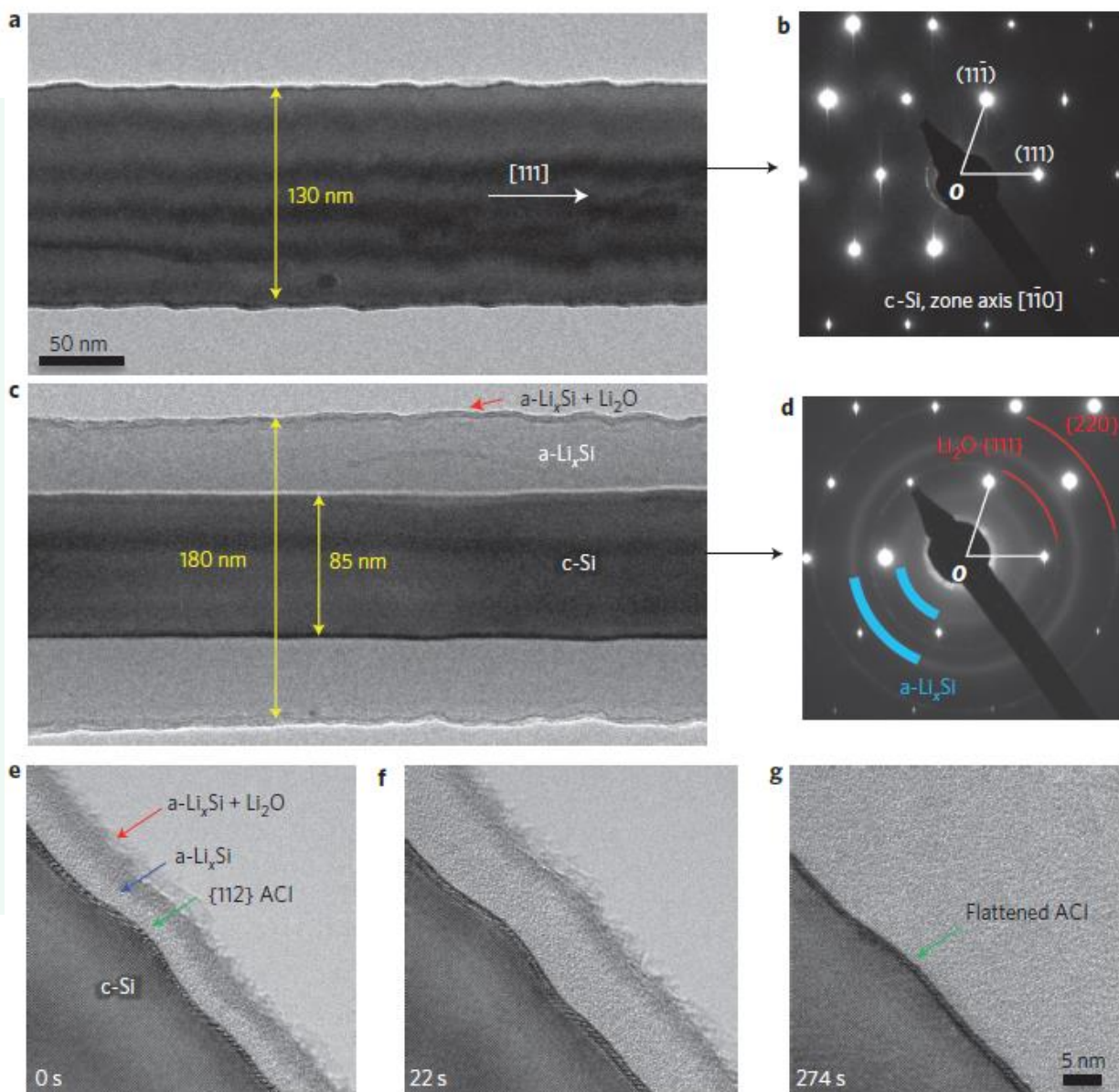
Huge Volumetric Change

Fast Disintegration of Particles

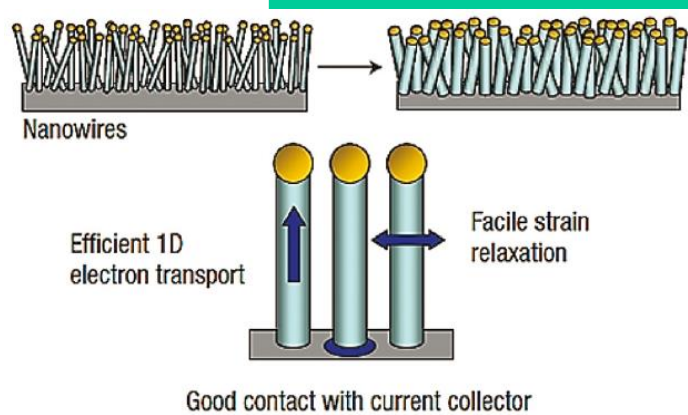
- The initial structure of electrode cannot remain during cycle.
- The electrode having huge volumetric expansion cannot work in a secondary battery.

硅负极锂化机理

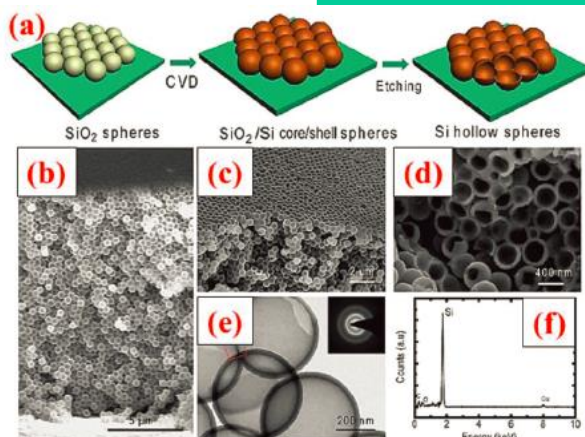
1. 无定形 Li_xSi 首先形成
2. 无定形/晶体界面随着锂化进程的进行不断向内部发展
3. 晶体硅不同原子面方向锂化速度不同，产生的应力也各向异形
4. 锂化动力学受界面迁移的控制，该迁移通过涉及侧向的壁架机制发生密集的 $\{111\}$ 原子平面上的运动。



纳米化

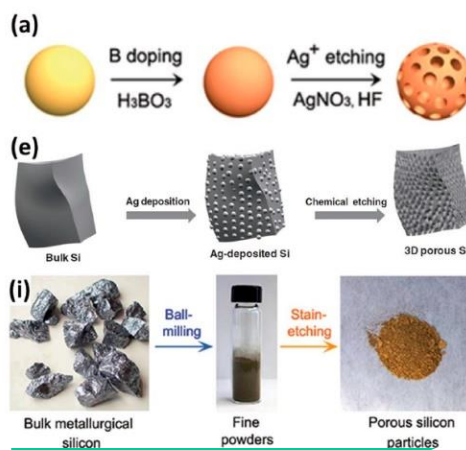


Nat. Nanotechnol., 2008, 3

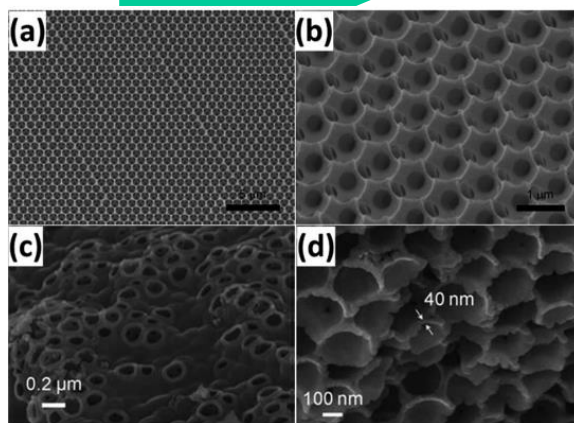


Nano Lett., 2011, 11

多孔化

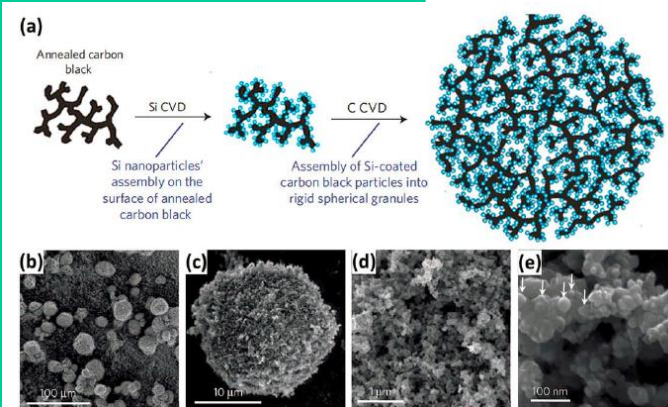


Nano Lett., 2014, 14

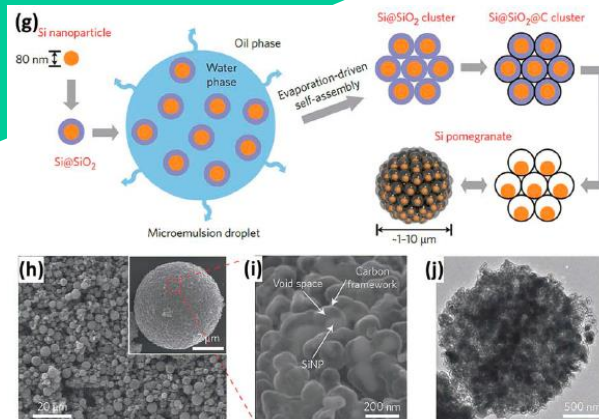


Angew. Chem., Int. Ed., 2008, 47

与碳复合



Nat. Mater., 2010, 9

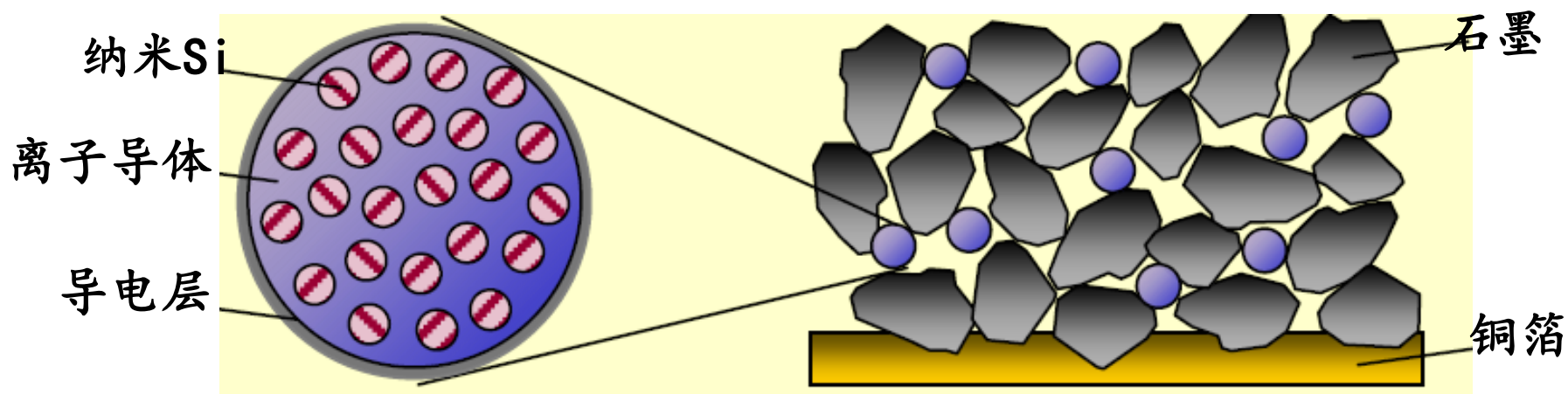


Nat. Nanotechnol., 2014, 9 6

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硅基材料与高结晶度碳的复合负极材料

发明了硅基材料与石墨、高结晶度碳的复合电极材料（JP 5046302和JP 5600559等）。基于该发明等，日立集团在**世界上首次实现了硅基锂离子电池的产业化**，并被**苹果、三星**等公司的手机所采用。

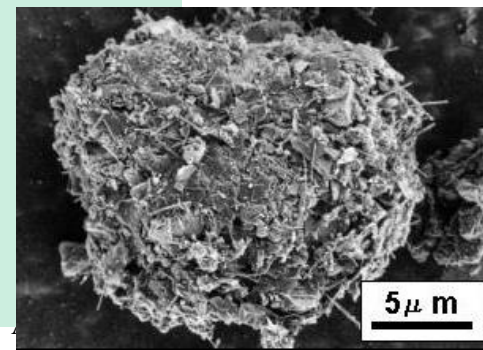


From homepage of Hitachi Maxell Ltd.

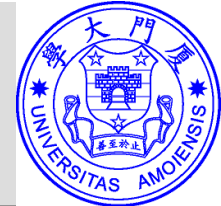
发明的要点:

In 2006

- 1) Si 粒子包碳;
- 2) 把 Si@C 或 SiX@C 与石墨混合;
- 3) Si 粒子的尺寸远小于石墨。



硅基电池的循环特性

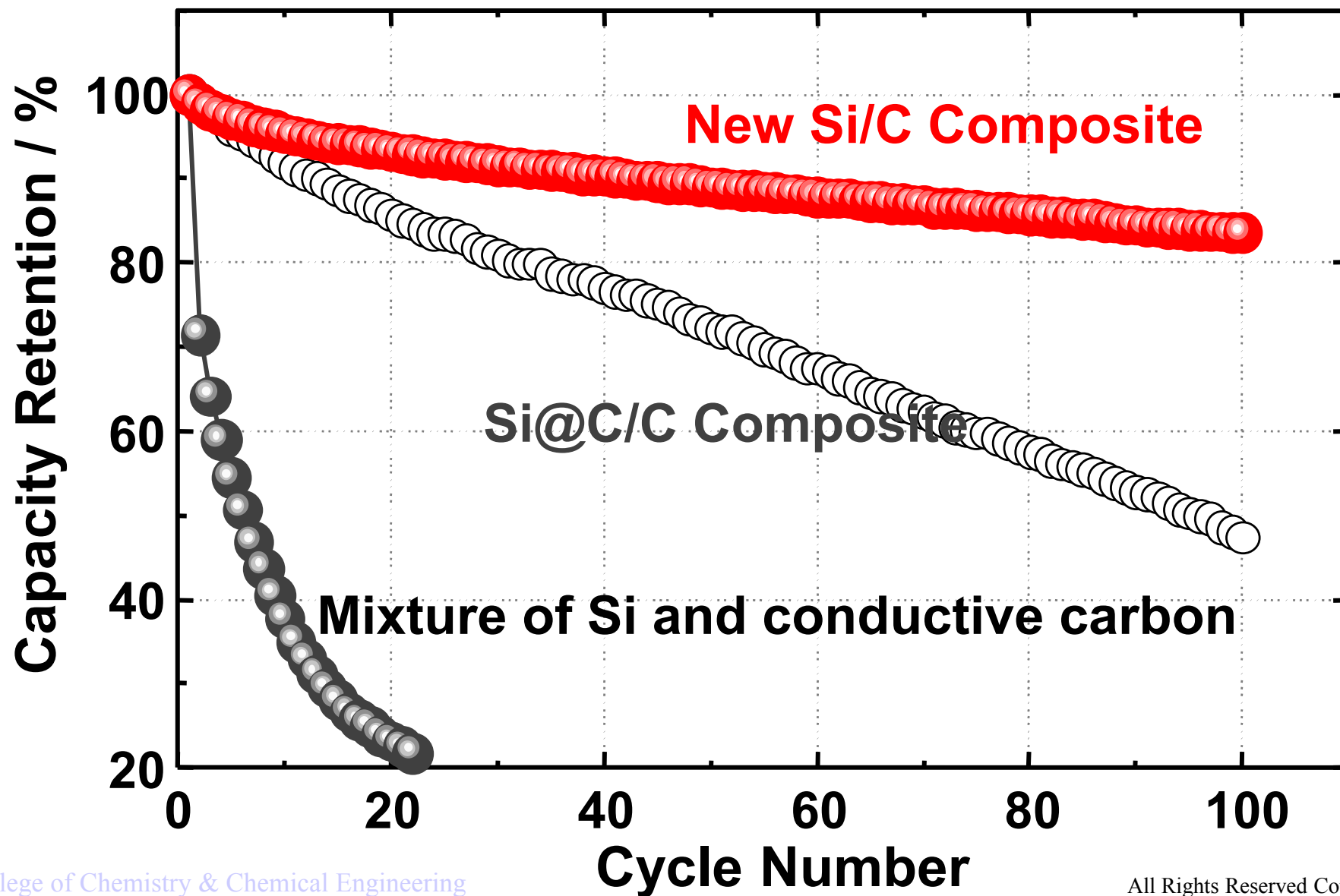


200mAh Card Type Cell

Cathode: M-LCO

Current: 0.5C

Temp.: 23°C



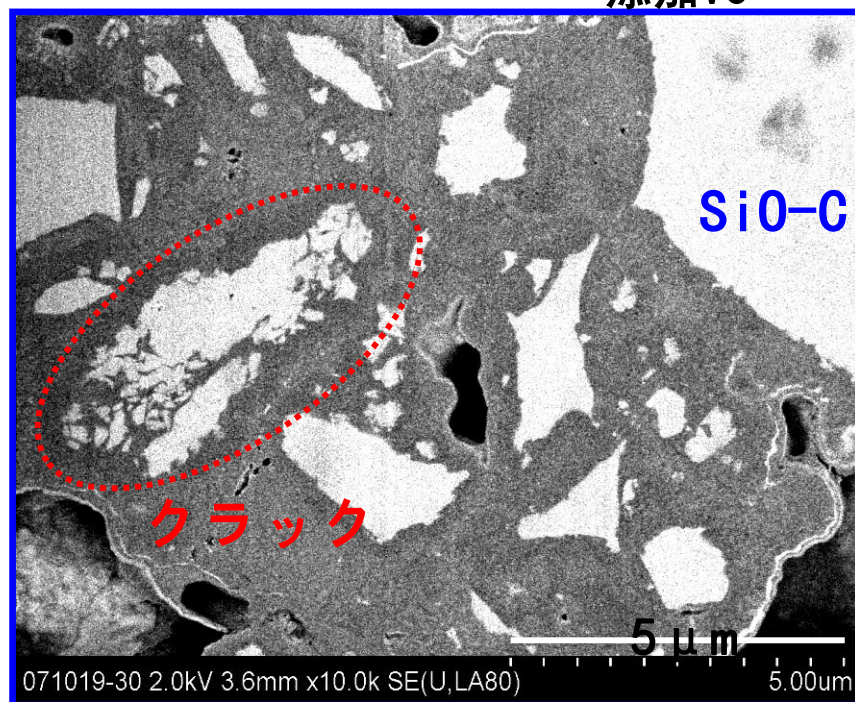
2007年の工作

SiO-C微粉化

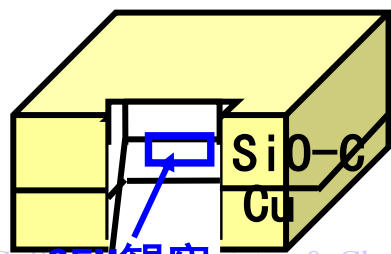


充电

添加VC



After 10Cycle



硅基负极产业化

➤ 2021年8月，成功搭建占地450m²硅基复合负极材料中试产线，日产30公斤；



中试线剪彩、投产



项目签约仪式

2021年11月，与杰瑞股份完成签约仪式，拉开正式合作序幕；



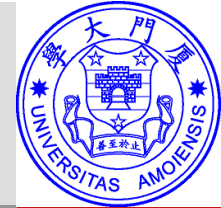
氧化亚硅成品



中试产线一角

锂离子电池的电解液

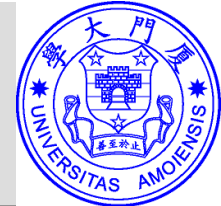
正负极是电池的基因，电解液是锂离子电池的血液，其组成与锂离子电池的健康状态(即性能)具有非常重要的关系。



Li^+ 的载体/电池充放过程中的呼吸作用

→ 80%以上的问题与电解液有关

	材料	体积变化 (%)
正极	LCO	1-3%
	NCM333	0%
	LFP	6%
	LMO	7%
	LTO	<1%
负极	Carbon	10%
	Sn基	30-100%
	Si	300%
	SiO-C	200%



要求

- 宽电位窗/宽使用温度范围
- 高离子电导度 (特别是 Li^+)
- 高电化学稳定性
- 特性改良 (功能化)

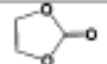
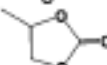
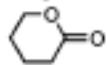
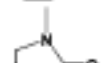
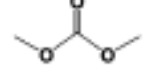
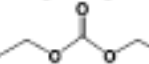
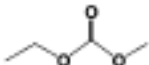
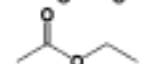
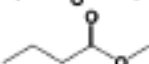
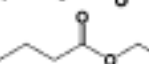
组成

- 溶 剂：稳定性、低粘度、高诱电率
- 电解质盐：稳定性、高溶解性
- 添加剂：电池特性改善

14 常用酯类溶剂的理化参数

碳酸酯类溶剂最重要！

Table 1. Organic Carbonates and Esters as Electrolyte Solvents

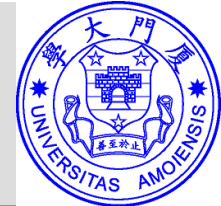
Solvent	Structure	M. Wt	T _m /°C	T _b /°C	η/cP 25 °C	ε 25 °C	Dipole Moment/debye	T _d /°C	d/gcm ⁻³ , 25 °C
EC		88	36.4	248	1.90, (40 °C)	89.78	4.61	160	1.321
PC		102	-48.8	242	2.53	64.92	4.81	132	1.200
BC		116	-53	240	3.2	53	4.23	199	1.199
γBL		86	-43.5	204	1.73	39	4.23	81	1.057
γVL		100	-31	208	2.0	34	4.29	110	1.17
NMO		101	15	270	2.5	78	4.52	18	1.063
DMC		90	4.6	91	0.59 (20 °C)	3.107	0.76	31	0.969
DEC		118	-74.3 ^a	126	0.75	2.805	0.96	11	0.898
EMC		104	-53	110	0.65	2.958	0.89	19	0.878
EA		88	-84	77	0.45	6.02	-3	0.902	
MB		102	-84	102	0.6				
EB		116	-93	120	0.71				

高诱电率

低粘度

^a The mp of DEC recorded in various literature sources (books, papers, commercial catalogs) has been -43 °C, which was corrected by a very recent measurement (ref 50e). This widespread error of 30° seems to stem from a single source in 1921, which was then registered by *Beilstein Handbuch* and escaped detection for approximately eight decades.

作业：写出EC、MEC、DMC的构造式，说明这些溶剂的特点



15 常用醚类溶剂的理化参数

Table 2. Organic Ethers as Electrolyte Solvents

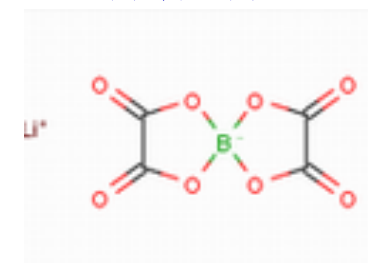
Solvent	Structure	M. Wt	T _m / °C	T _b / °C	η/cP 25 °C	ε 25 °C	Dipole Moment/debye	T _f / °C	d/gcm ⁻³ , 25 °C
DMM		76	-105	41	0.33	2.7	2.41	-17	0.86
DME		90	-58	84	0.46	7.2	1.15	0	0.86
THF		72	-109	66	0.46	7.4	1.7	-17	0.88
2-Me-THF		86	-137	80	0.47	6.2	1.6	-11	0.85
1,3-DL		74	-95	78	0.59	7.1	1.25	1	1.06
4-Me-1,3-DL		88	-125	85	0.60	6.8	1.43	-2	0.983
2-Me-1,3-DL		88			0.54	4.39			

抗氧化性较低！

锂金属电池常用，但锂离子电池少用

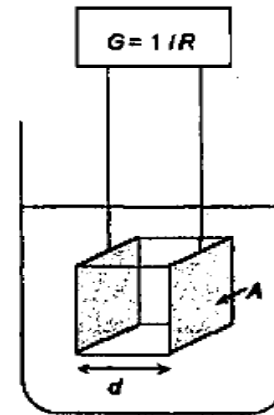
- LiBF_4 : 高温性能比较好, 但是价格昂贵和溶解度比较低;
- LiPF_6 : 综合性能比较好, 缺点是易吸水水解;
- LiAsF_6 : 综合性能比较好, 但是毒性太大;
- LiClO_4 : 综合性能比较好, 但是强氧化性导致安全性不高;
- LiBOB : 高温性能比较好, 尤其能抑制溶剂对负极的插入破坏, 但是溶解度太低。 二草酸硼酸锂、Lithium bis(oxalate)borate

强酸盐、大尺寸的阴离子



17 离子导电性

$$\sigma = \sum_i n_i u_i Z_i e$$



$$\theta = \frac{d}{A} \quad \theta : \text{セル定数}$$

$$R = \rho \theta \quad R : \text{抵抗} \\ \rho : \text{抵抗率}$$

$$G = \frac{\kappa}{\theta} \quad G : \text{電気伝導度} \\ \kappa : \text{電気伝導率}$$

$$\Lambda = \frac{1000 \kappa}{c}$$

c : モル濃度 (mol dm^{-3})

κ : 電気伝導率 (S cm^{-1})

Λ : モル伝導率 ($\text{S cm}^2 \text{mol}^{-1}$)

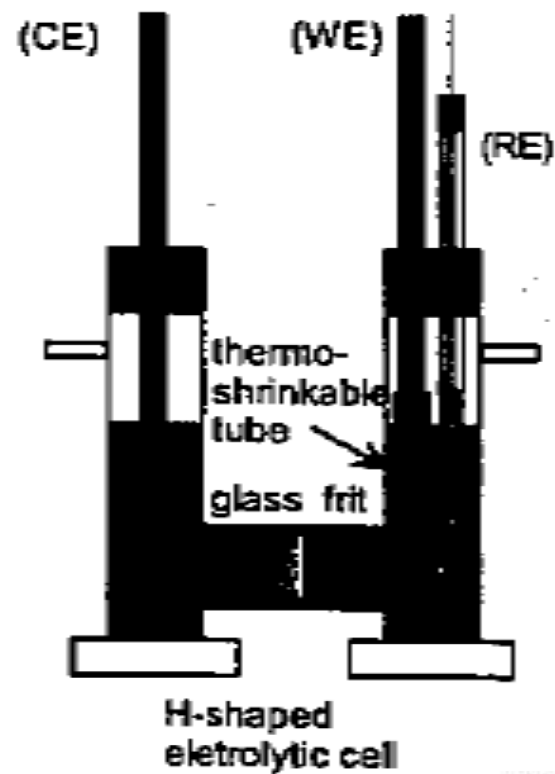
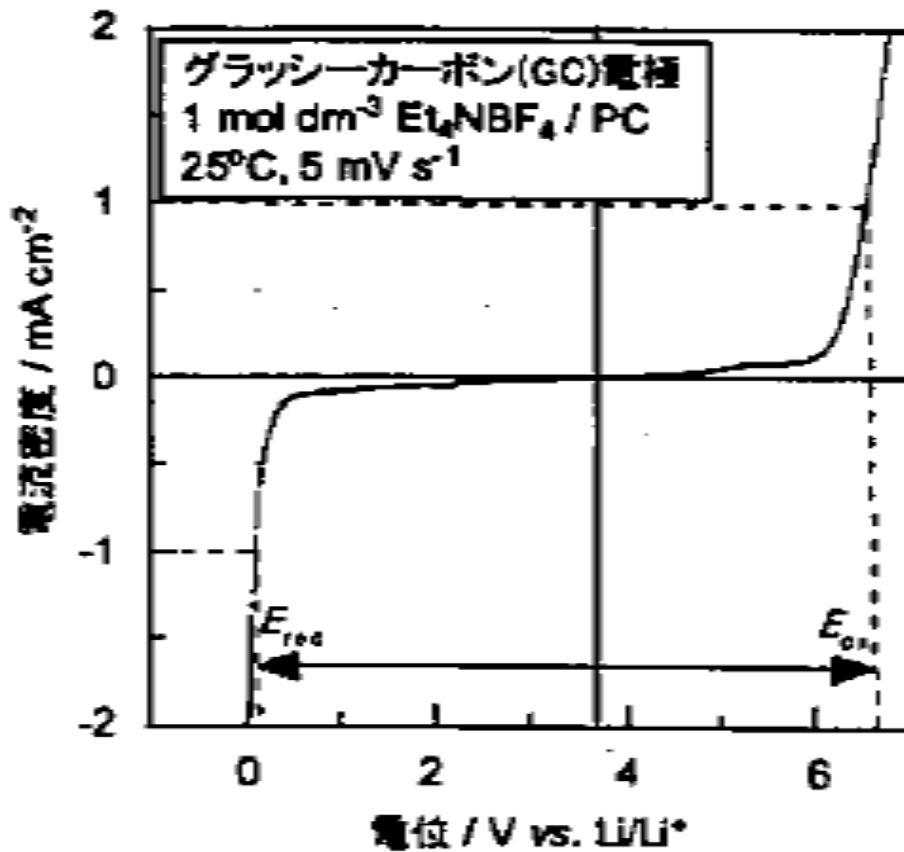
$$\sigma = f(m, x, T)$$

$$\sigma = AT^{1/2} e^{-B/R(T-T_0)}$$

粘度、温度、誘電率、濃度...

18 电位窗的测定

Linear Sweep Voltammetry (LSV)



常用玻炭电极

常用参比电极的电极电势

参比电极	温度 (°C)	电极电势 (V)
Hg Hg ₂ Cl ₂ 饱和 KCl	25	0.245
Hg Hg ₂ Cl ₂ 1N KCl	25	0.280 1
Hg Hg ₂ Cl ₂ 0.1N KCl	25	0.333 7
Ag AgCl 饱和 KCl	25	0.198 1
Ag AgCl 0.1N HCl	25	0.287
Hg HgO 0.1N NaOH	25	0.164

19 溶剂的电位窗

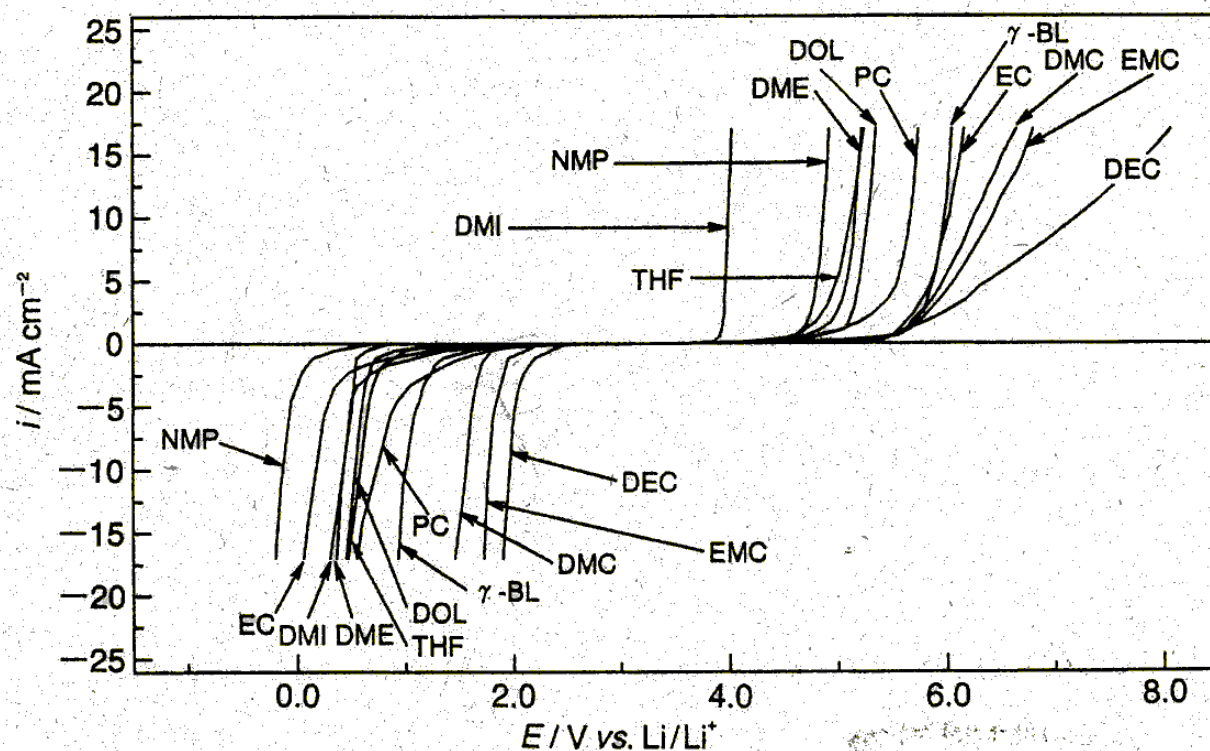


図 4 TBAP 含有溶液の i - E 曲線. TBAP(過塩素酸テトラブチルアンモニウム)の濃度: 0.1 mol dm^{-3} , 掃引速度: 2 mV s^{-1} . PC: プロピレンカーボネート, EC: エチレンカーボネート, γ -BL: γ -ブチロラクトン, DMC: ジメチルカーボネート, EMC: エチルメチルカーボネート, DEC: ジエチルカーボネート, DOL: 1,3-ジオキソラン, THF: テトラヒドロフラン, DME: 1,2-ジメトキシエタン, NMP: 1-メチル-2-ピロリジノン, DMI: 1,3-ジメチル-2-イミダゾリジノン.

TBAP: 四丁基高氯酸铵

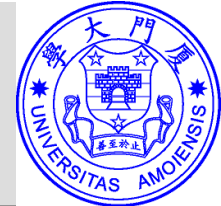


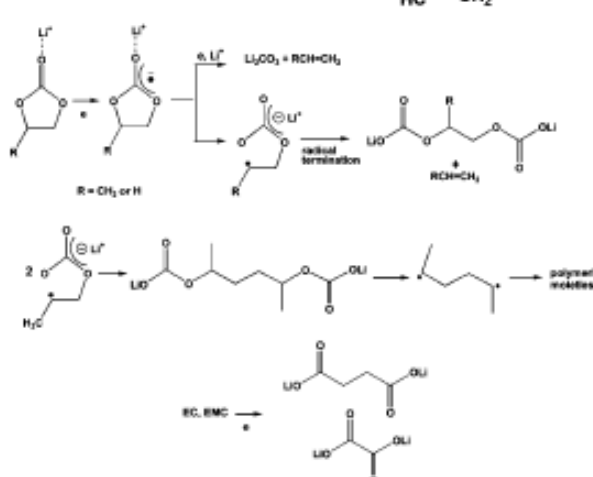
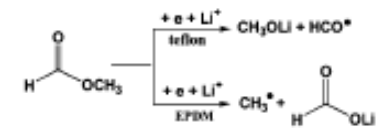
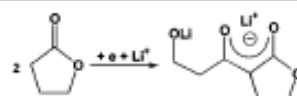
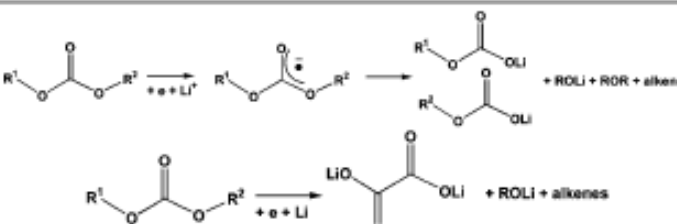
Table 5. Electrochemical Stability of Electrolyte Solvents: Nonactive Electrodes

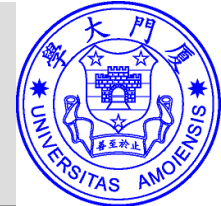
solvent	salt/conc (M)	working electrode	E_a^a	E_c^b	ref
PC	Et ₄ NBF ₄ /0.65	GC	6.6		130, 131
	none ^c	Pt	5.0	~1.0	74
	Bu ₄ NPF ₆	Ni		0.5	212
	LiClO ₄ /0.1	Au, Pt		1.0–1.2	214
	LiClO ₄ /0.5	porous Pt	4.0		305
	LiClO ₄	Pt	4.7		177
	LiClO ₄	Au	5.5		306
	LiAsF ₆	Pt	4.8		177
	Et ₄ NBF ₄ /0.65	GC	6.2		130, 131
EC	Bu ₄ NPF ₆	Ni		0.9	212
	LiClO ₄ /0.1	Au, Pt		1.36	214
	Et ₄ NBF ₄ /0.65	GC	6.7		130, 131
DMC	LiClO ₄ /0.1	Au, Pt		1.32	214
	LiPF ₆ /1.0	GC	6.3		76
	LiF	GC	5.0		312
	Et ₄ NBF ₄ /0.65	GC	6.7		130, 131
DEC	LiClO ₄ /0.1	Au, Pt		1.32	214
	Et ₄ NBF ₄ /0.65	GC	6.7		130, 131
EMC	LiPF ₆ /1.0	GC	6.7		76
	LiAsF ₆ /0.5	Au, Ag		1.25	208
γBL	Et ₄ NBF ₄ /0.65	GC	5.2		130, 131
THF	LiClO ₄	Pt	4.2		177
	LiAsF ₆ /1.0	GC	4.25 (0.1)		214
	none ^c	Pt	4.0	< -2.0	74
	LiAsF ₆ /1.0	GC	4.2		64
	LiAsF ₆	Pt	4.2		177
	LiClO ₄	Pt	4.1		177
	LiAsF ₆ /1.0	GC	4.15 (0.1)		214
	LiAsF ₆ /1.0	GC	4.2		64
	LiAsF ₆	Pt	4.1		177
DME	Et ₄ NBF ₄ /0.65	GC	5.1		130, 131
	LiClO ₄	Pt	4.5		177
	LiAsF ₆	Pt	4.5		177

较低!

^{a,b} Anodic and cathodic limits, potential referred to Li⁺/Li. ^c The salt-free condition was realized via an ultramicroelectrode technique.

Table 6. Reductive Decompositions on Carbonaceous Anodes

Solvents	Reactions	Refs.
PC, EC:	$\text{PC} + 2\text{e} + 2\text{Li} \longrightarrow \text{Li}_2\text{CO}_3 + \text{H}_3\text{C}-\text{HC}=\text{CH}_2$  $\text{EC, BMC} \xrightarrow{\text{e}^-} \text{radical anion}$	237 55, 104, 125, 249, 250, 274, 277, 281 262 278
MF:		249
γBL:		123
Linear Carbonates:		108, 123, 214, 271, 279 284



B. Electrolyte Salts and Other Components

Salt	LiClO_4: $\text{LiClO}_4 + ne + 2n \text{Li}^+ \rightleftharpoons \text{Li}_2\text{O} + \text{LiClO}_3, \text{LiClO}_2 + \text{LiCl}$
	LiAsF_6: $\text{LiAsF}_6 + 2e + 2 \text{Li}^+ \longrightarrow \text{AsF}_3 + 3\text{LiF}$ $\text{AsF}_3 + 2ne + 2 n \text{Li}^+ \longrightarrow \text{Li}_n\text{AsF}_{3-n} + n \text{Li}^+$
	LiPF_6: $\text{LiPF}_6 + 2e + 2\text{Li} \longrightarrow \text{PF}_3 + 3\text{LiF}$ $\text{LiPF}_6 \rightleftharpoons \text{LiF} + \text{PF}_5$ $\text{PF}_5 + \text{H}_2\text{O} \longrightarrow 2\text{HF} + \text{POF}_3$ $\text{PF}_5 + 2ne + 2 n \text{Li}^+ \longrightarrow \text{LiPF}_{6-n} + n \text{LiF}$
Contaminants	H_2O: $\text{H}_2\text{O} + e + \text{Li}^+ \longrightarrow \text{LiOH} + \text{H}_2$
	CO_2: $2\text{CO}_2 + 2e + 2\text{Li}^+ \longrightarrow \text{Li}_2\text{CO}_3 + \text{CO}$
	O_2: $\text{O}_2 + 4e + 4\text{Li}^+ \longrightarrow 2\text{Li}_2\text{O}$
Secondary Reactions	$\text{LiMF}_n + \text{H}_2\text{O} \longrightarrow \text{LiM}_x\text{O}_y + n\text{HF} + \text{MO}_2\text{F}$ $M = \text{P or B};$ $n = 4, 6$ $\begin{array}{c} \text{O} \\ \parallel \\ \text{R-O-C-OLi} \end{array} + \text{HF} \longrightarrow \text{LiF}$ $2 \begin{array}{c} \text{O} \\ \parallel \\ \text{R-O-C-OLi} \end{array} + \text{H}_2\text{O} \longrightarrow \text{Li}_2\text{CO}_3 + \text{CO}_2 + 2\text{ROH}$ $2\text{CO}_2 + 2e + 2\text{Li}^+ \longrightarrow 2\text{Li}_2\text{CO}_3 + \text{CO}$ $\text{Li}_2\text{CO}_3 + 2\text{HF} \longrightarrow \text{LiF} + \text{CO}_2 + \text{H}_2\text{O}$

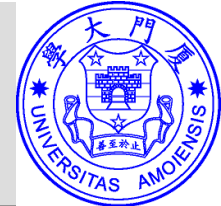
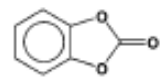
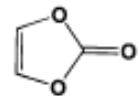
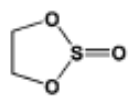


Table 9. Anode SEI Additives

Additive Structures	Host Electrolyte System	Remarks
CO_2	LiAsF ₆ /MF LiAsF ₆ /EC/DMC etc.	Non-effective for PC-suppression; internal pressure.
SO_2	LiClO ₄ /EC/DME LiAsF ₆ /MF LiAsF ₆ DMC, LiAsF ₆ /PC etc.	Effective for PC-suppression; anodic stability unknown; corrosive; safety hazard.
 12-crown-4	LiAsF ₆ /PC LiClO ₄ /EC/PC	Effective for PC-suppression; preferential solvation of Li ion; high concentration needed; toxic.
 18-crown-6	LiClO ₄ /EC/PC	Not as effective as 12-crown-4; high concentration needed; toxic.
 catecholic carbonate (CC)	LiPF ₆ /PC/DEC	Effective for PC-suppression; anodically stable.
 vinylene carbonate (VC)	LiIm/PC/EC/DMC LiPF ₆ /PC LiPF ₆ /DMC LiAsF ₆ /EC/DMC	Effective for PC-suppression; increases SEI stability at high temp.; anodically stable.
 ethylene sulfite (ES)	LiClO ₄ /PC	Effective for PC-suppression; anodically stable.

SEI: Solid electrolyte interphase

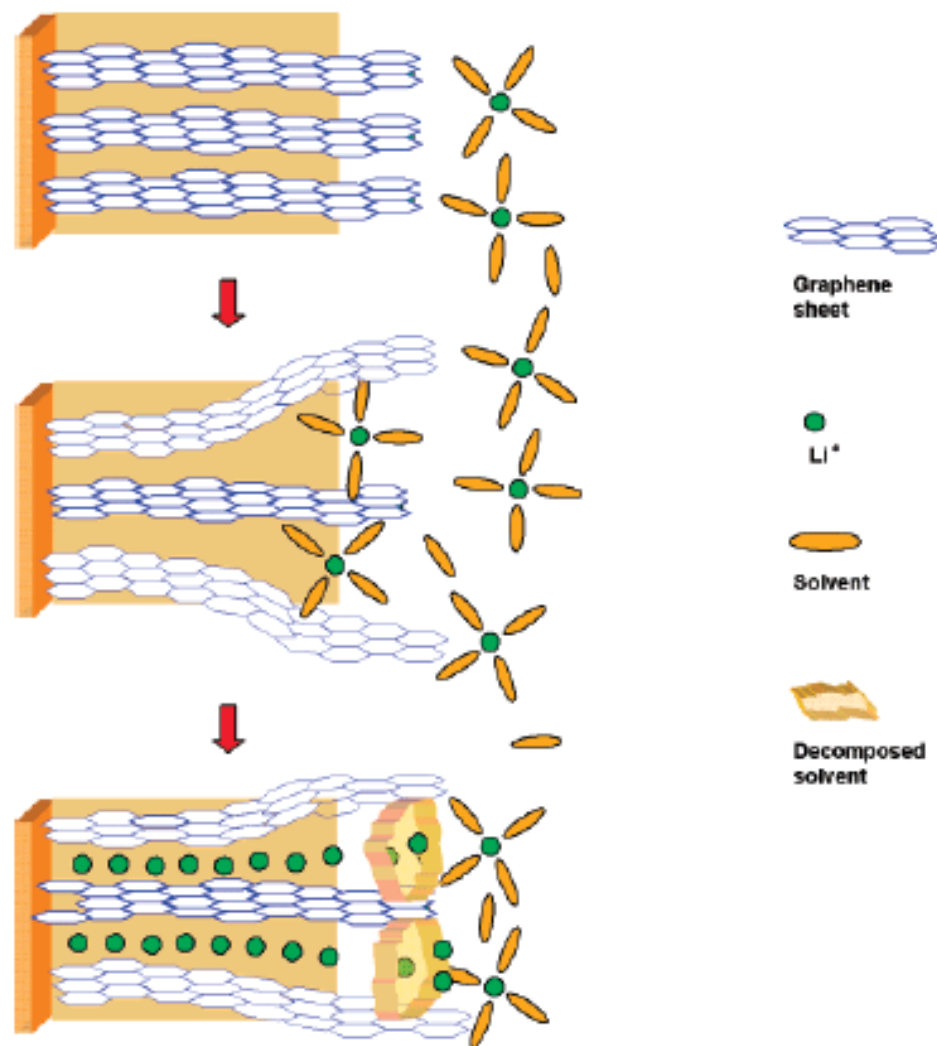


Figure 12. Schematic illustration of the SEI formation mechanism via the decomposition of $\text{Li(solv)}_x\text{C}_y$. Reconstructed based on ref 251.

常用溶剂的还原电位

溶 剂	还原电位
VC	1.20–1.30V
EC/DEC	1.0V: 溶剂化的 Li^+ 的嵌入 0.65V: 直接还原分解反应 EC/DEC比其他溶剂体系反应激烈
PC	1.0V: 溶剂化的 Li^+ 的嵌入

- Li^+ 的配位数:

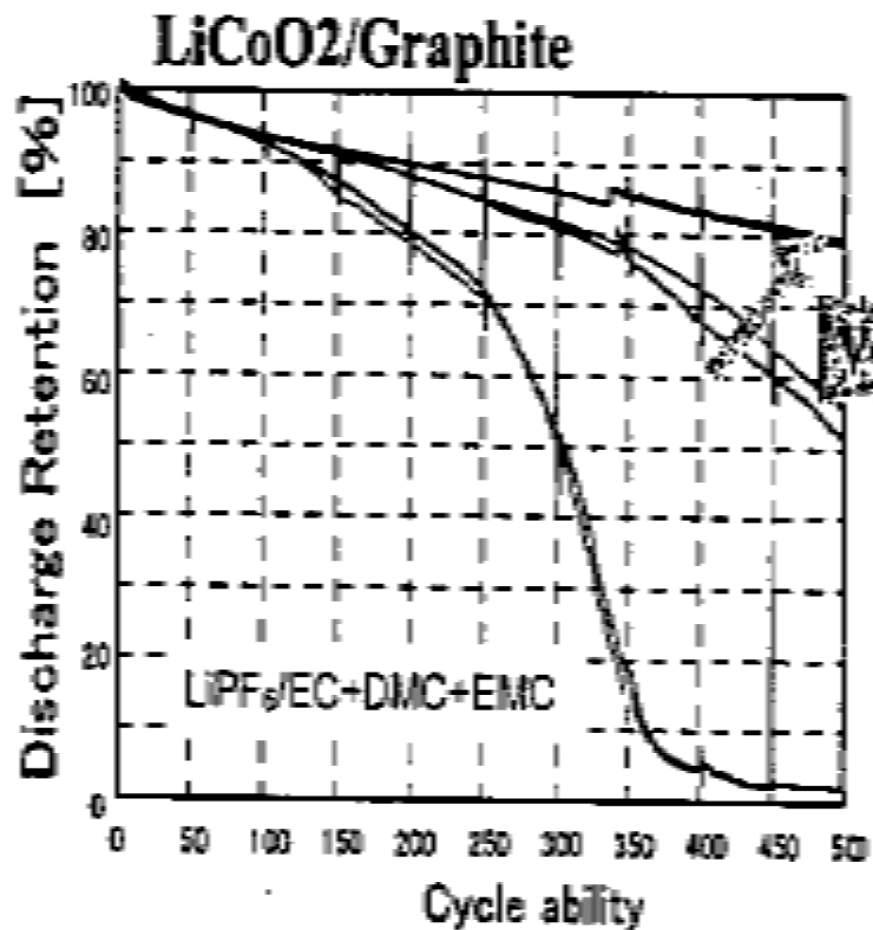
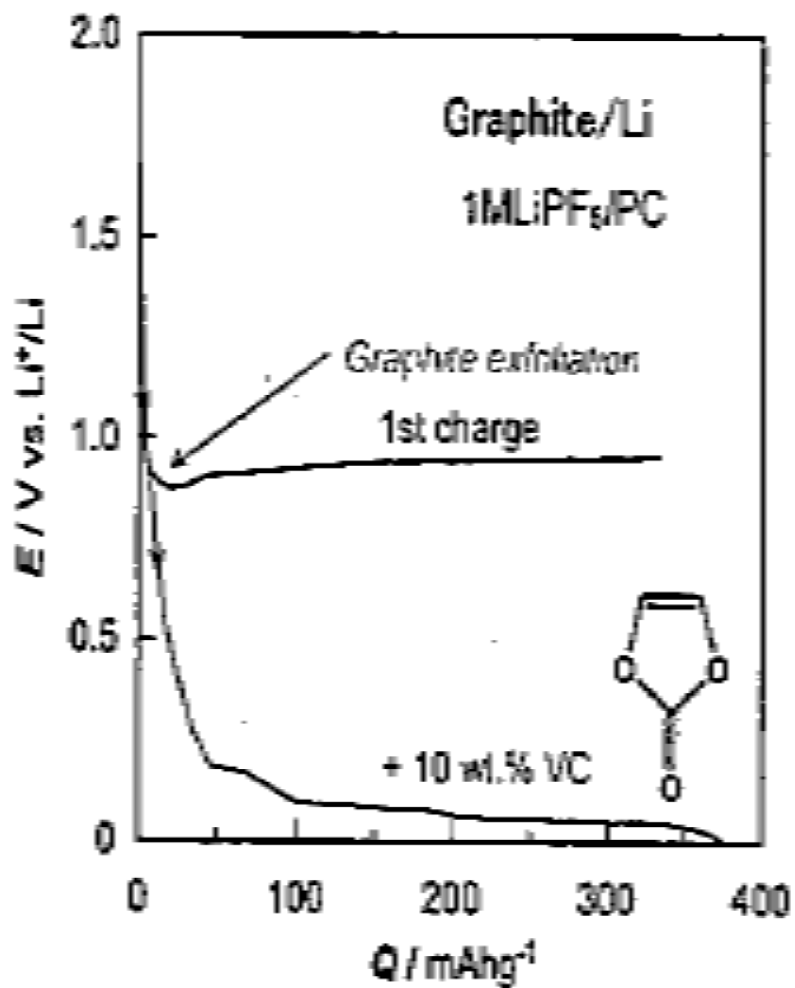
100%EC:

EC=4.6

EC/DEC=50/50:

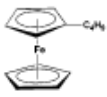
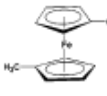
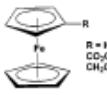
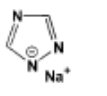
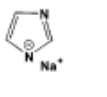
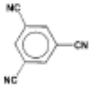
EC=3.1; DEC=1.1

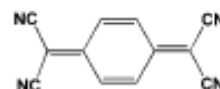
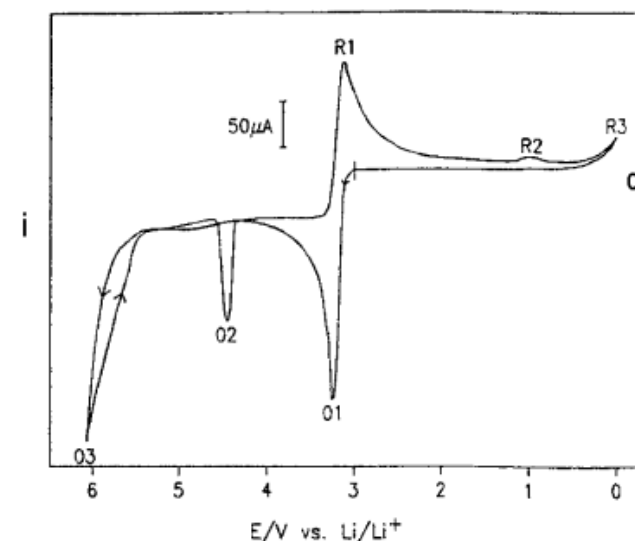
负极SEI形成提高电池循环特性



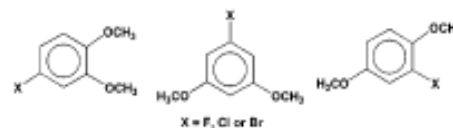
高HOMO

Table 11. Cathode Surface Layer Additives: Overcharge Protection

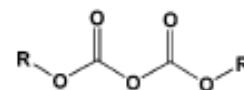
Additive Structures	Host Electrolyte System	Type	E_{onset}/V	
I/I_2	LiAsF ₆ /THF	Redox	3.2~3.5	4
 <i>n</i> -butylferrocene	LiAsF ₆ /EC/2-Me-THF	Rodox	3.18	
 1,1'-dimethylferrocene		Rodox	~ 3.0	
 ferrocene derivatives R = H, Bu, COCH₃, COC₆H₅, CO₂CH₃, CONH₂, CH₂N(CH₃)₂, CH₂OH, CH₂(CH₃)₂, CH₂OCH₃	LiAsF ₆ /EC/PC	Rodox	3.09~3.55	
 1,2,4-Triazole, Na salt	LiIm/PEO	Rodox	3.1	
 imidazole, Na salt	LiIm/PEO	Rodox	2.5	
	LiIm/PEO	Rodox	2.9	



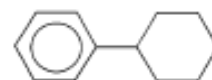
tetracyanoquinodimethane (TCNQ)



substituted benzenes



pyrocarbonate



cyclohexylbenzene

LiIm/PEO Rodox 3.5

LiPF₆/PC/DMC Rodox 3.93~4.27

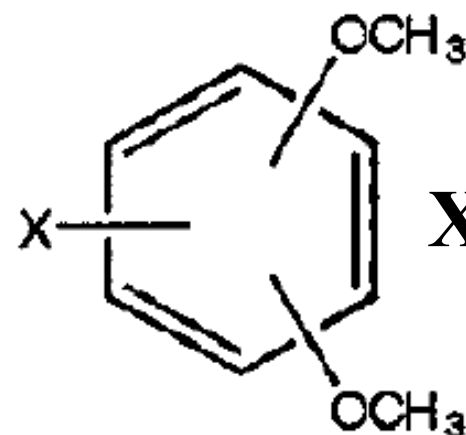
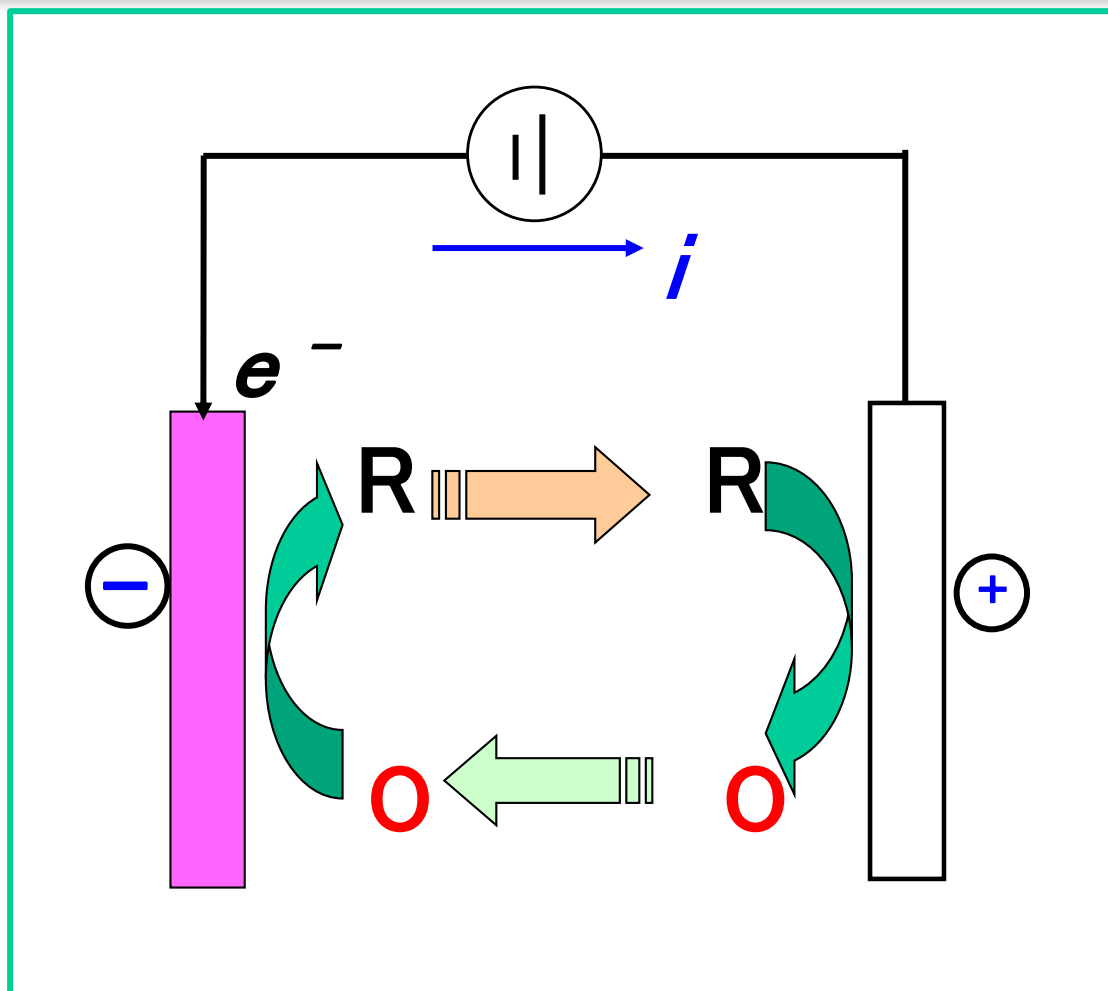
Gas/CID^a > 4.0

Gas/CID > 4.0

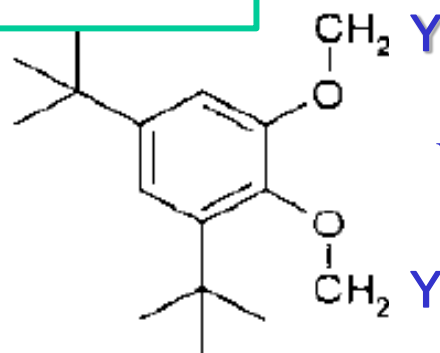
LIB的正常工作电位下, 正极添加剂不参加反应!

作业：为什么？

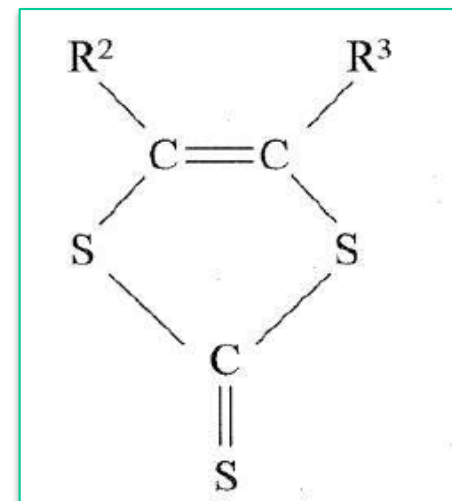
过充电保护 ---氧化还原对 (Redox Shuttle)



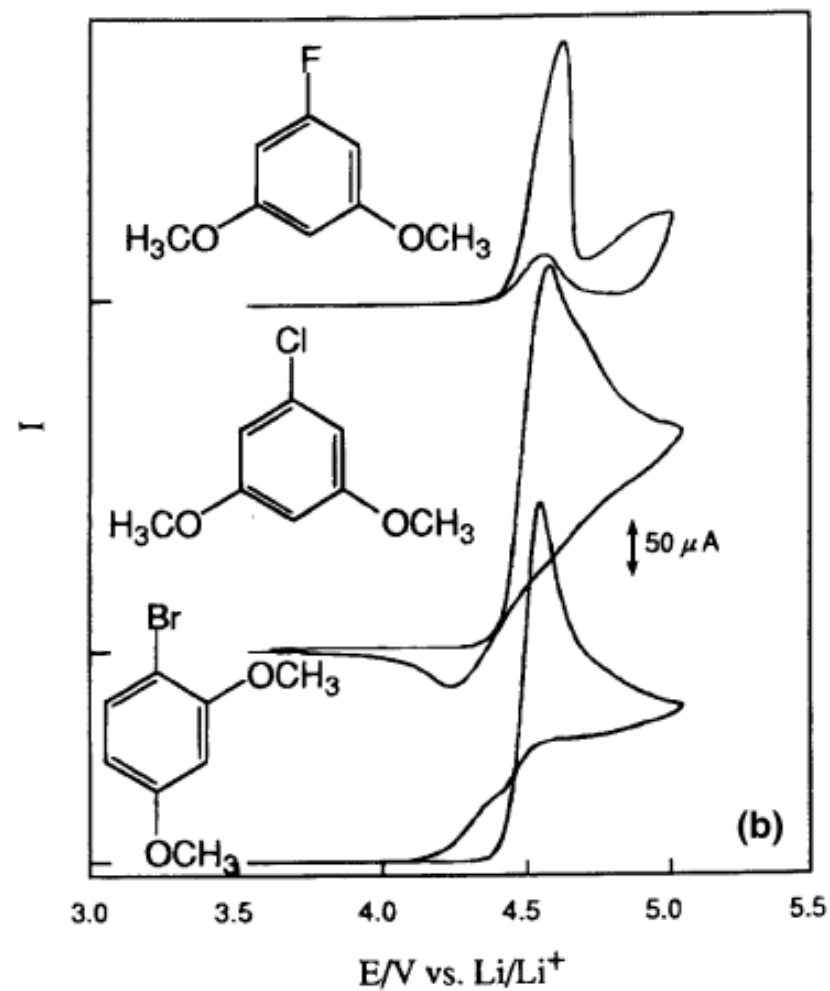
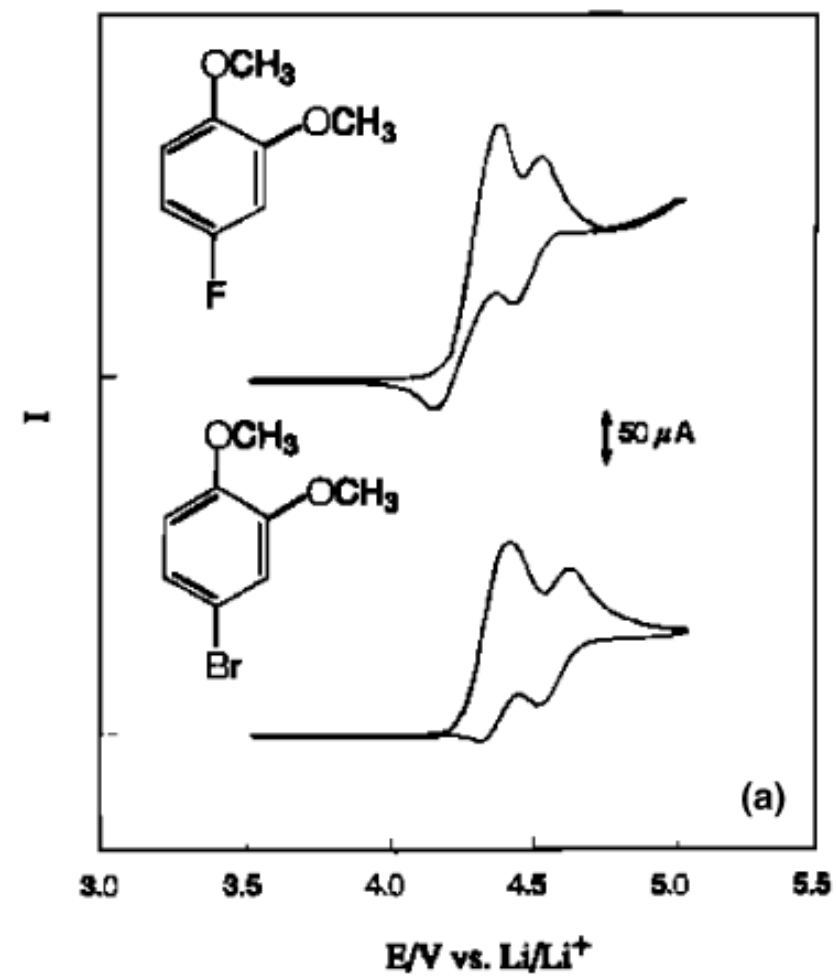
$\text{X} = \text{Cl}, \text{F}, \text{Br}$



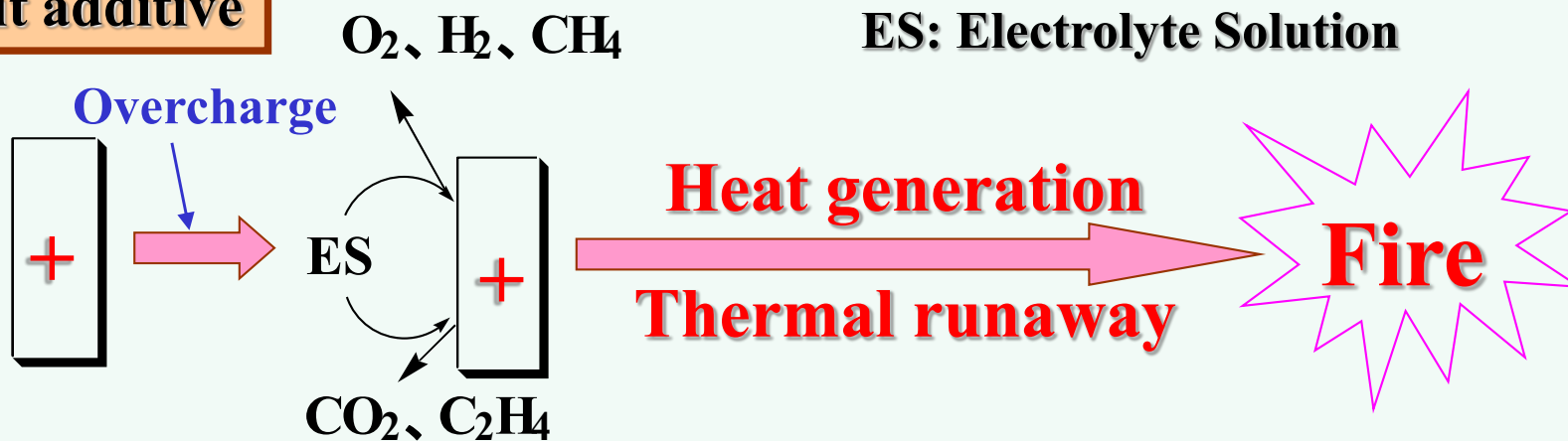
$\text{Y} = \text{H}, \text{F}$



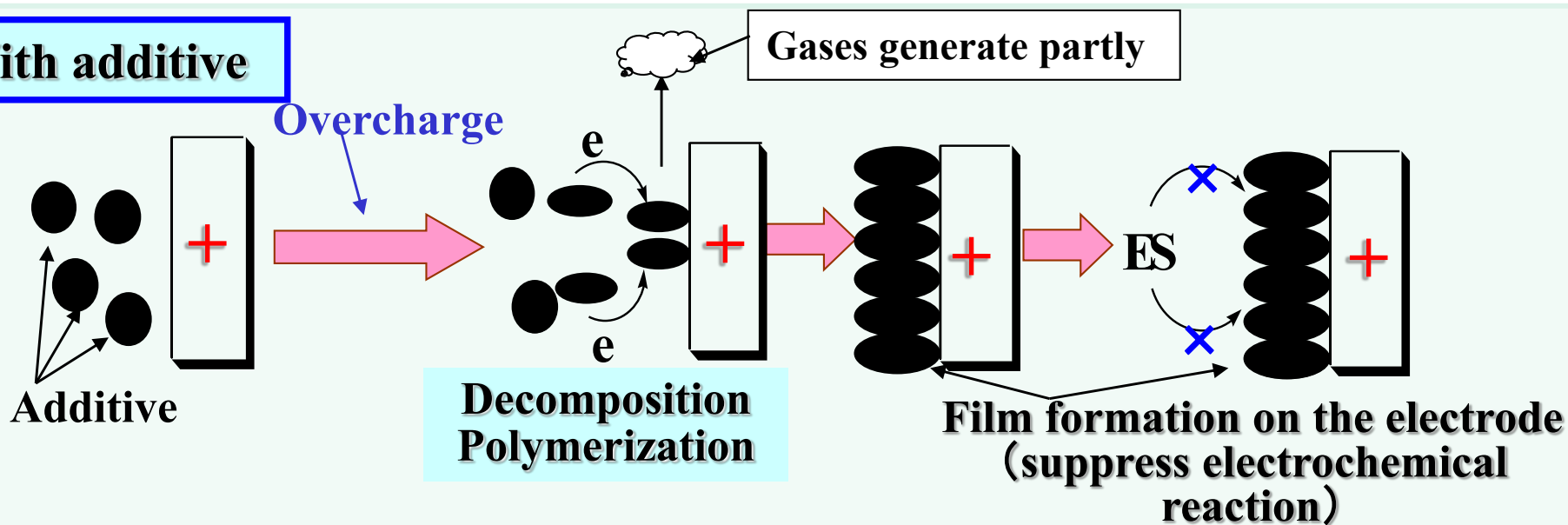
防止过充电效果有限!



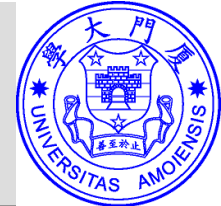
Without additive

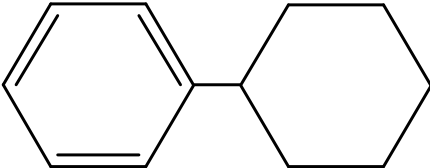
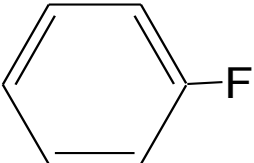
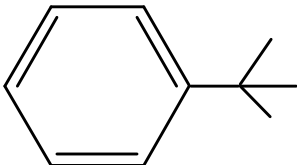
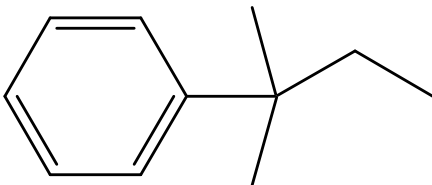
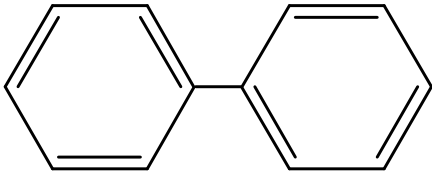


With additive

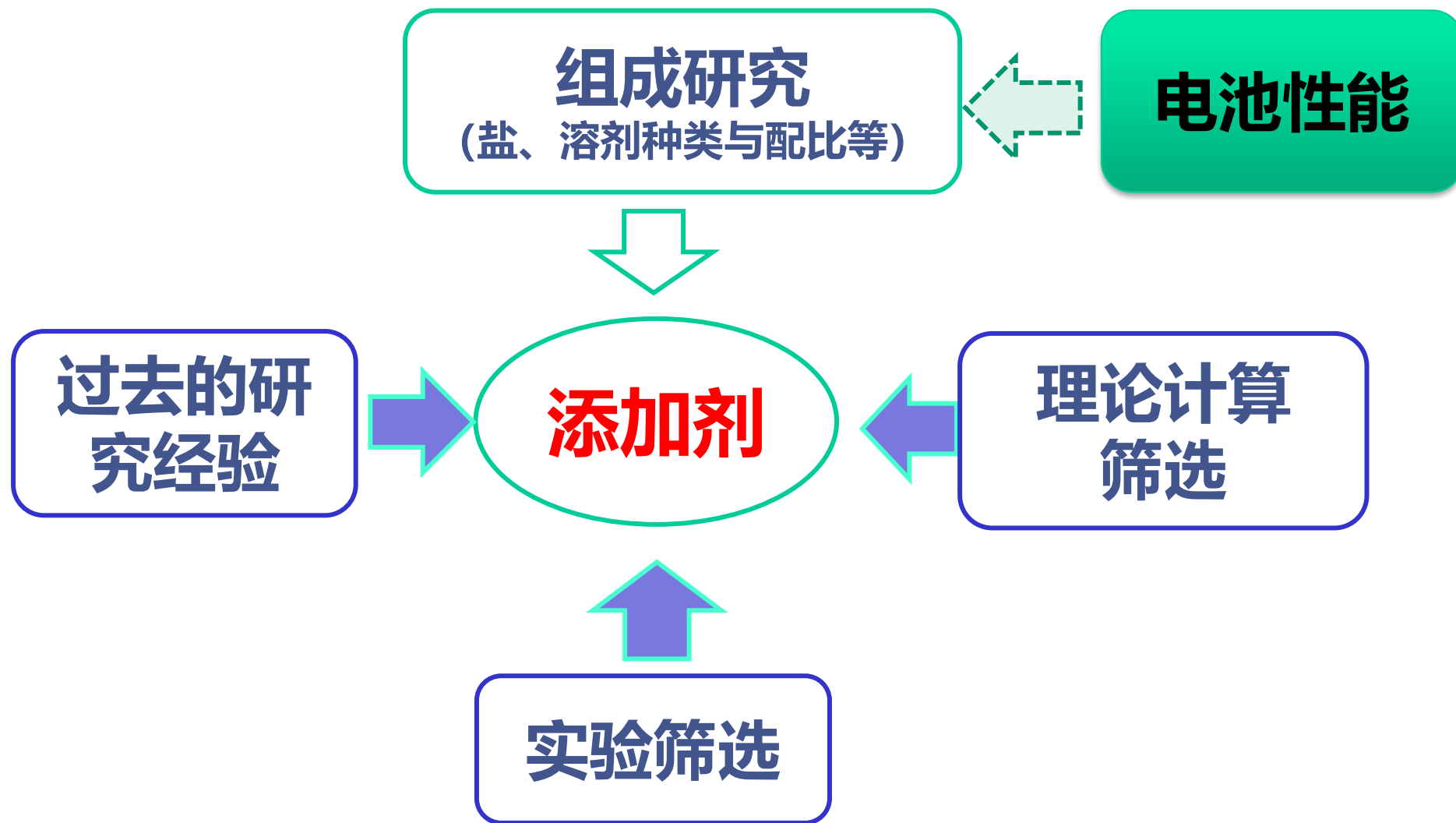
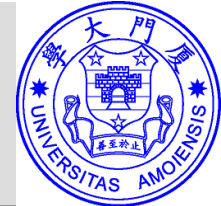


31 常用防止过充电用添加剂的氧化电位

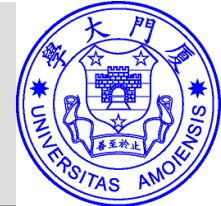


添加剂	结构式	反应电压 (V) (vs. Li/Li ⁺)
CB		4.62
FB		5.10
<i>t</i> -BB		5.10
<i>t</i> -AB		5.05
BP		4.54

IVSM法测定 (WE=Pt 线)

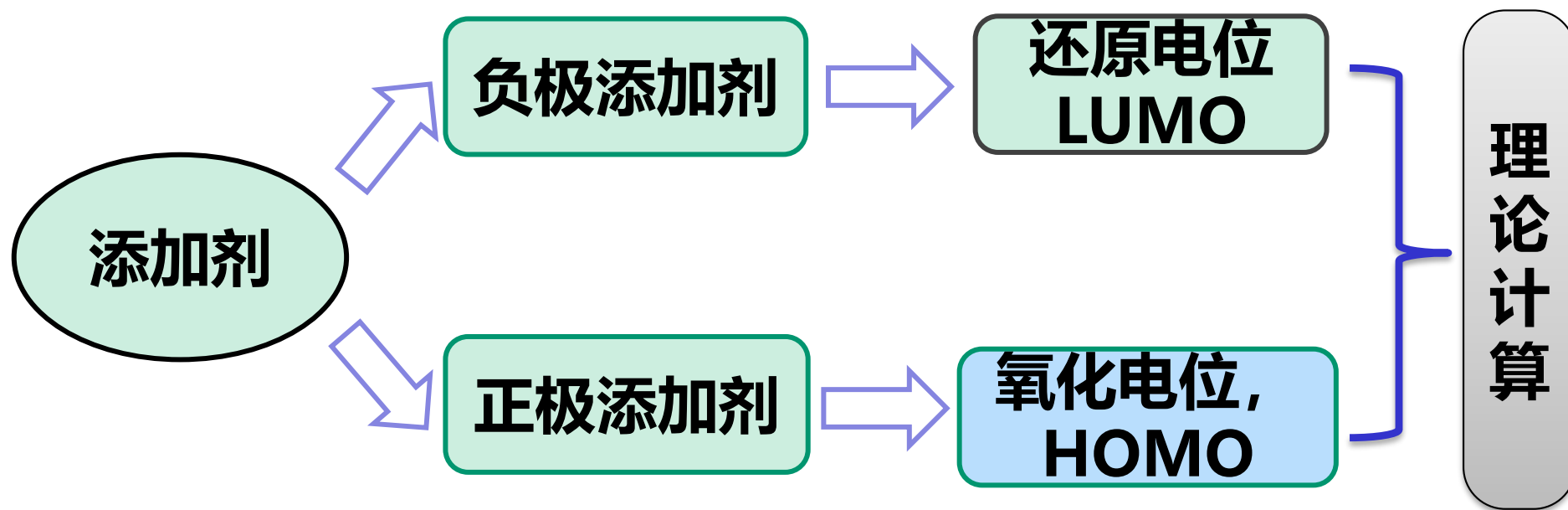


33 分子轨道计算



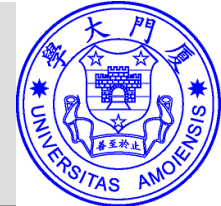
HOMO : 最高已占轨道, 越低, 抗氧化性越好。

LUMO : 最低未占轨道, 越高, 耐还原性越好。



热力学稳定性: 低HOMO (难氧化) 、高LUMO (难还原)

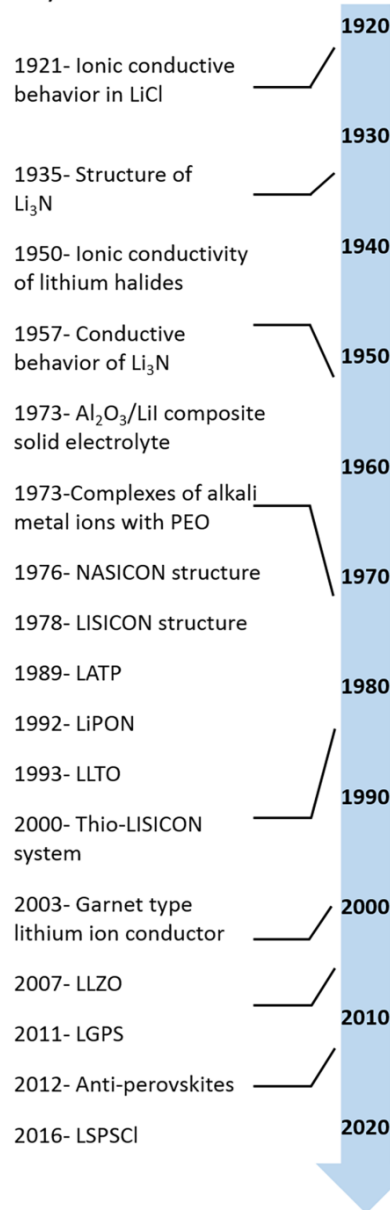
34 市场亟待解决的课题



- ☆ 高性能电池用功能电解液
- ☆ 高电压电池用电解液
- ☆ Si基负极材料用电解液
- ☆ 极限环境（超低温、超高温）电解液

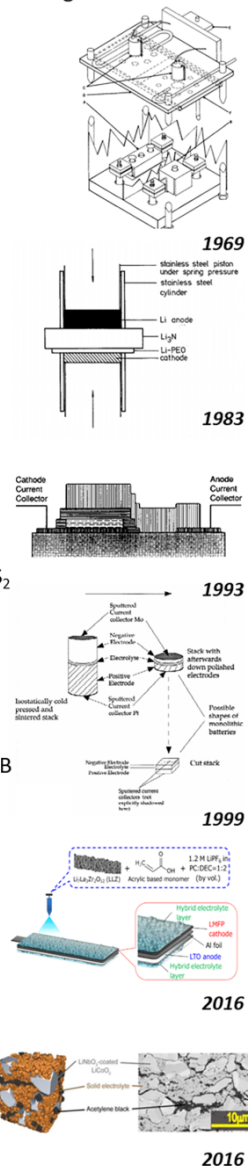
35 固体电解质：液体→固体

Key materials



All-Solid-State Battery technologies

Configurations of ASSBs



历史与分类

Chem. Rev.2019,DOI10.1021/acs.chemrev.9b0026

固态电解质

无机固态电解质

有机固态电解质

氧化物固
态电解质

硫化
物固
态电
解质

聚
醚体
系

聚碳酸
酯体系

其他
体系

钙钛矿型
LiPON
LISICON型
NASICON型
石榴石型
(Garnet)

$\text{Li}_2\text{S}-\text{P}_2\text{S}_5$

PEO
及其
衍生
物

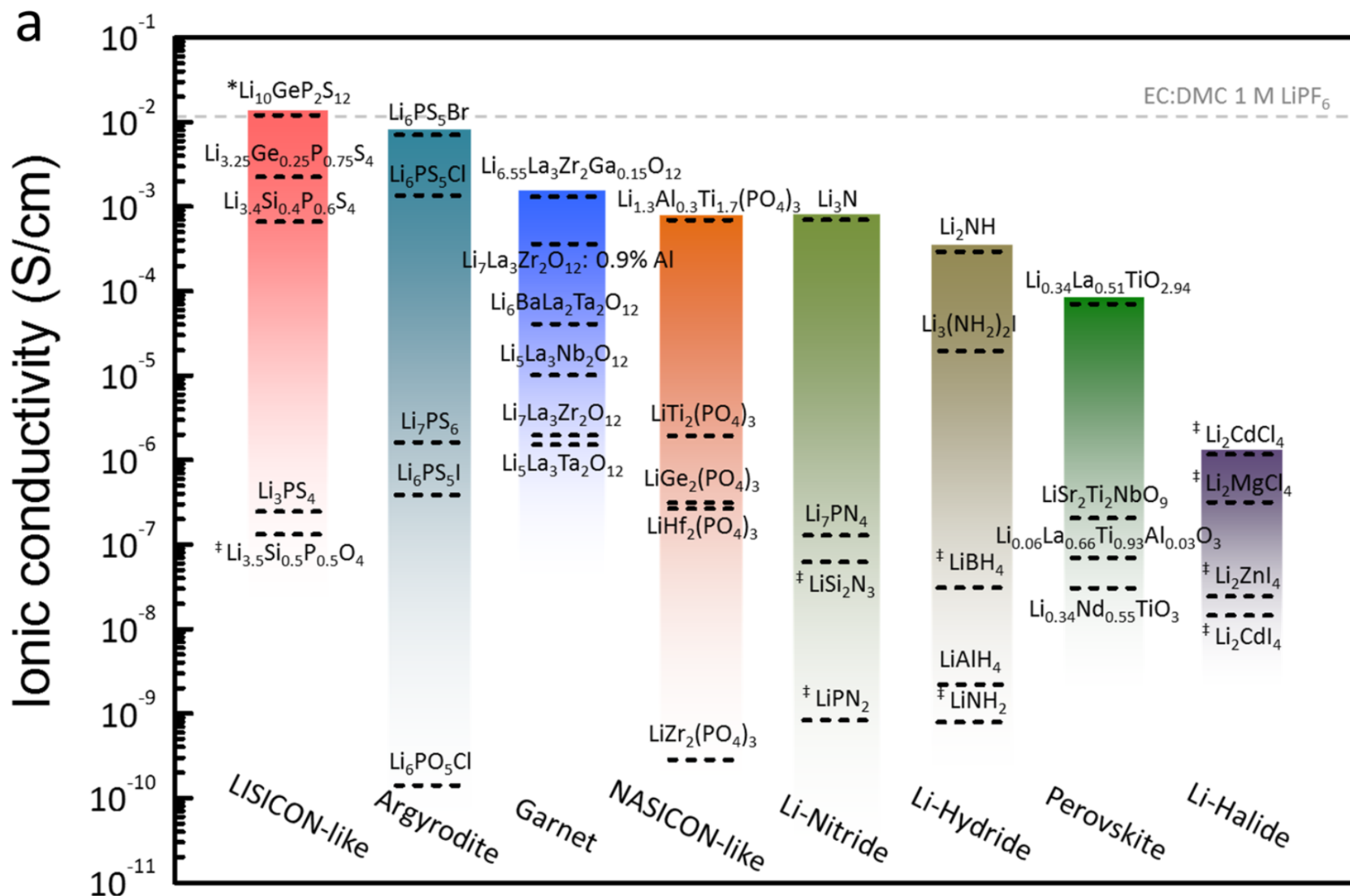
PEC,
PPC等

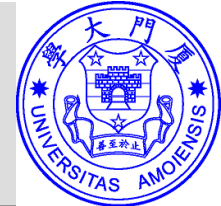
聚硅
氧烷
体系;
聚丙
烯腈
体系

无机固态电解质特点与区别：

氧化物固态电解质	优势	劣势
钙钛矿型	合成简单，晶粒电导率大 10^{-3} S/cm。	晶界电阻大， Ti^{4+} 易被锂金属还原成 Ti^{3+} 。
LISICON型	高温下具有较高电导率	室温电导率低，对空气中的水和 CO_2 敏感。
NASICON型	室温电导率高 10^{-3} S/cm，化学稳定性。	晶界电阻大， Ti^{4+} 易被锂金属还原成 Ti^{3+} 。
石榴石型	电化学稳定性好，室温电导率较高 10^{-4} S/cm。	对空气中的水和 CO_2 敏感。

无机固态电解质离子电导率对比图

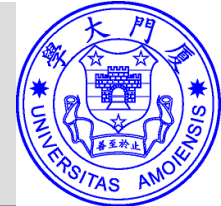




定义： 聚合物电解质是以聚合物为基体，由强极性的聚合物和金属盐通过Lewis酸-碱络合络合反应形成的在固态下形成的具有离子导电性的功能高分子材料。

研究历史：

- ✓1973年，Fenton最早发现聚环氧乙烯（PEO）可以与碱金属形成聚合物-盐络合物；
- ✓1975年，Wright首先报道了该体系的络合物具有优异离子导电性。
- ✓1979年，Armand报道了PEO的碱金属盐络合物体系具有良好的成膜性能，可用作锂电池的电解质（室温导电率为 10^{-5}S/cm ）。



聚合物电解质的特点：

- ✓ 具有高分子材料的柔顺性、良好的成膜型，粘弹性、稳定性、和良好的电化学稳定性
- ✓ 抑制锂枝晶的生长
- ✓ 增强电池对循环过程中电极体积变化的承受能力
- ✓ 减低电极材料与液体电解质的反应性
- ✓ 更高的安全性
- ✓ 更高的灵活性和制作一体性

40 聚合物电解质



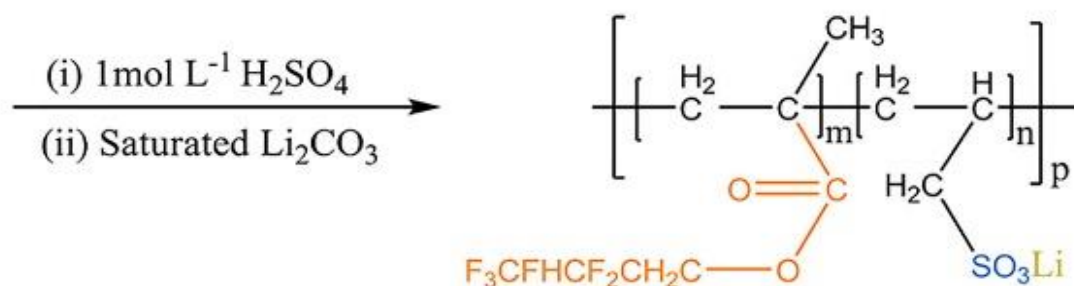
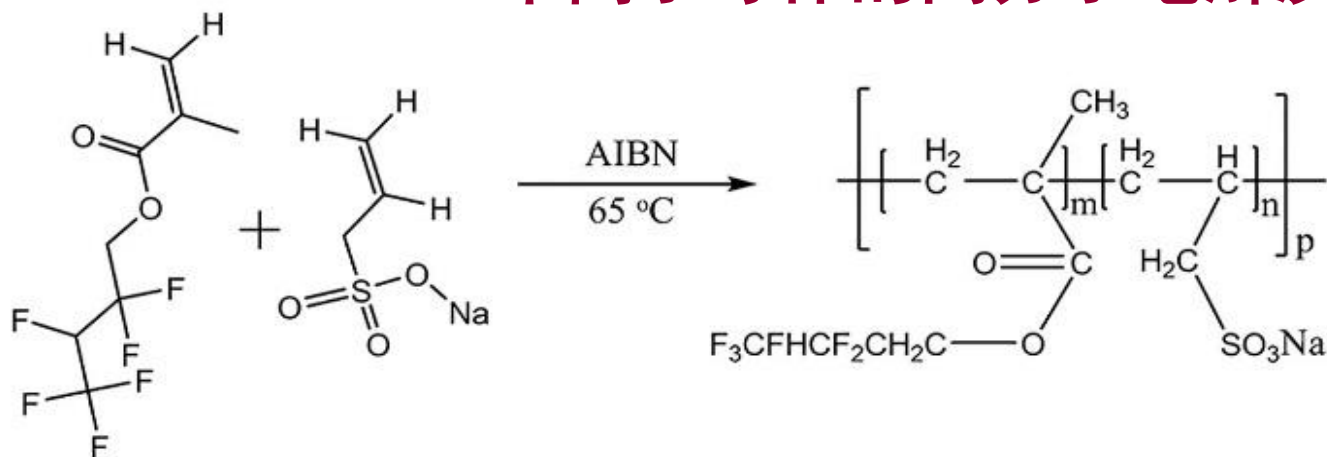
常用的聚合物电解质的宿主材料及其化学式

中文名称	英文名称	缩写	$T_g/^\circ\text{C}$	$T_m/^\circ\text{C}$
聚氧乙烯	poly(ethylene oxide)	PEO	64	65
聚氧丙烯	Poly(propylene oxide)	PPO	60	①
聚丙烯腈	poly(acrylonitrile)	PAN	25	317
聚氯乙烯	Poly(vinyl chloride)	PVC	82	①
聚偏氟乙烯	Poly(vinylidene fluoride)	PVDF	40	171

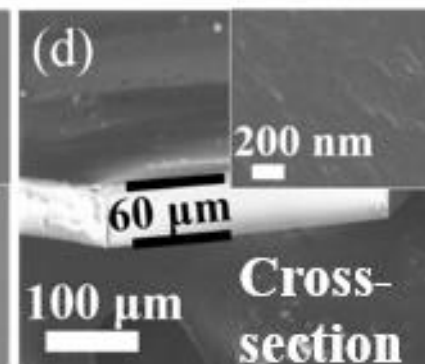
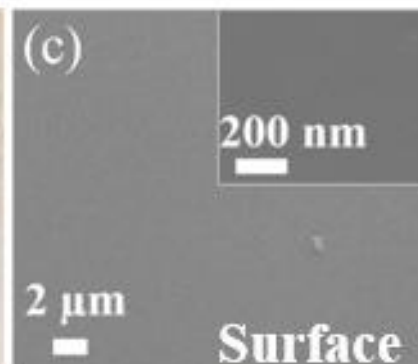
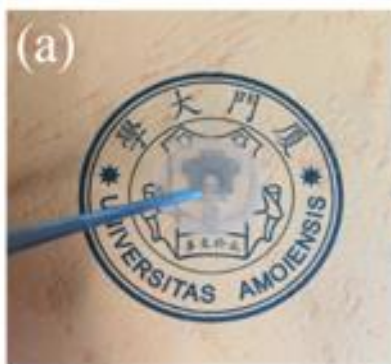
①为无定型 (amorphous) 聚合物

注： T_g 玻璃化转变温度， T_m 熔点

单离子导体的高分子电解质隔膜



甲基丙烯酸六氟丁酯-
烯丙基磺酸锂共聚物
P(HFMA-co-ASLi)



高能辐射接枝

预辐射PVDF粉体



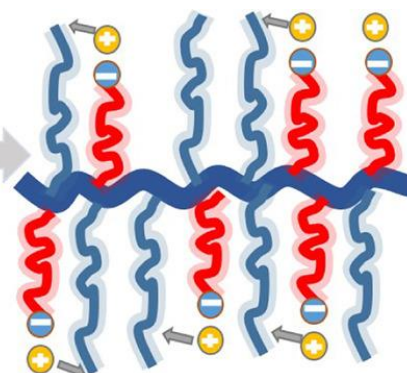
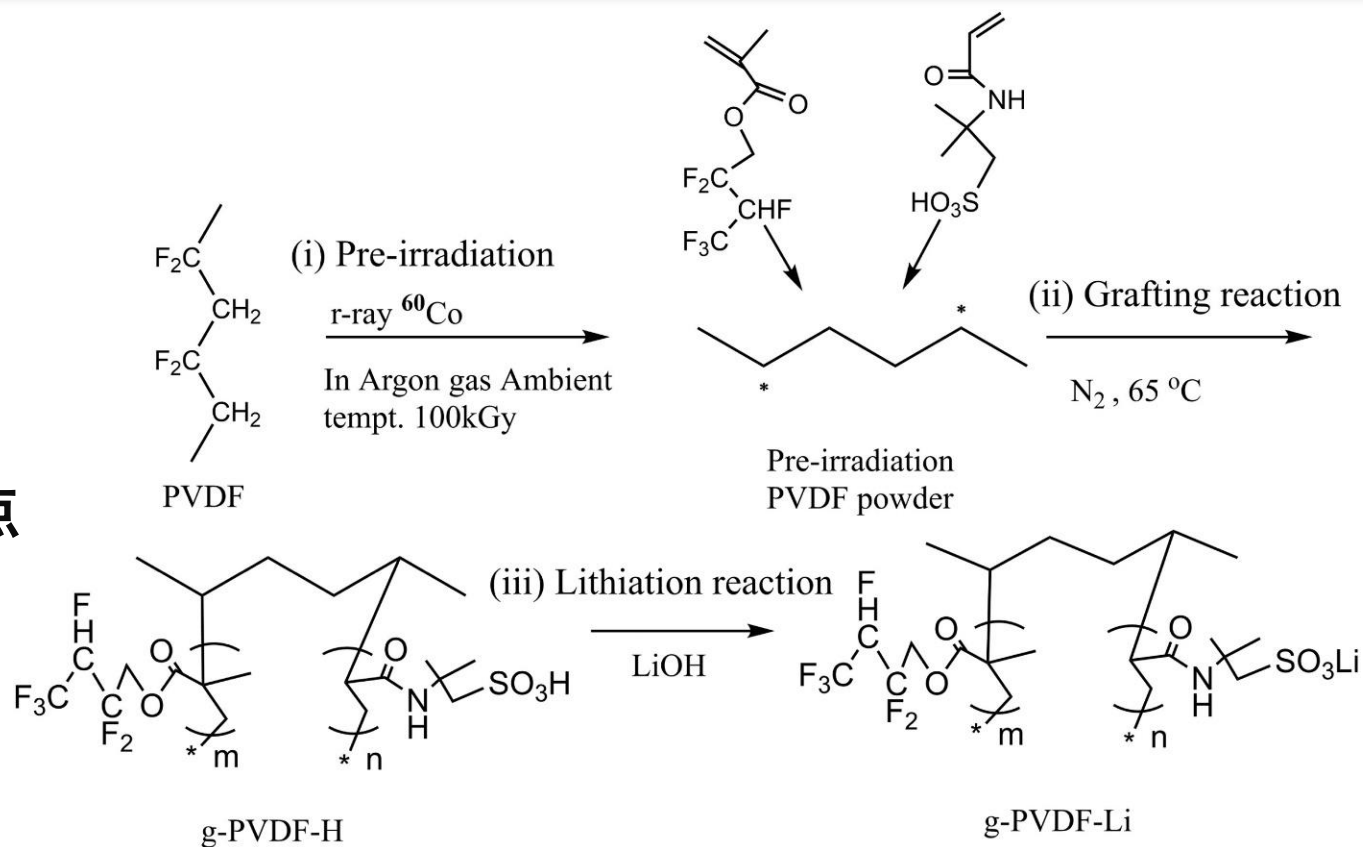
均匀的高能自由基活性位点



位点上分散生长
接枝AMPS、HFMA



梳状单离子导体
g-PVDF-Li



Li-ion source (AMPSLi)



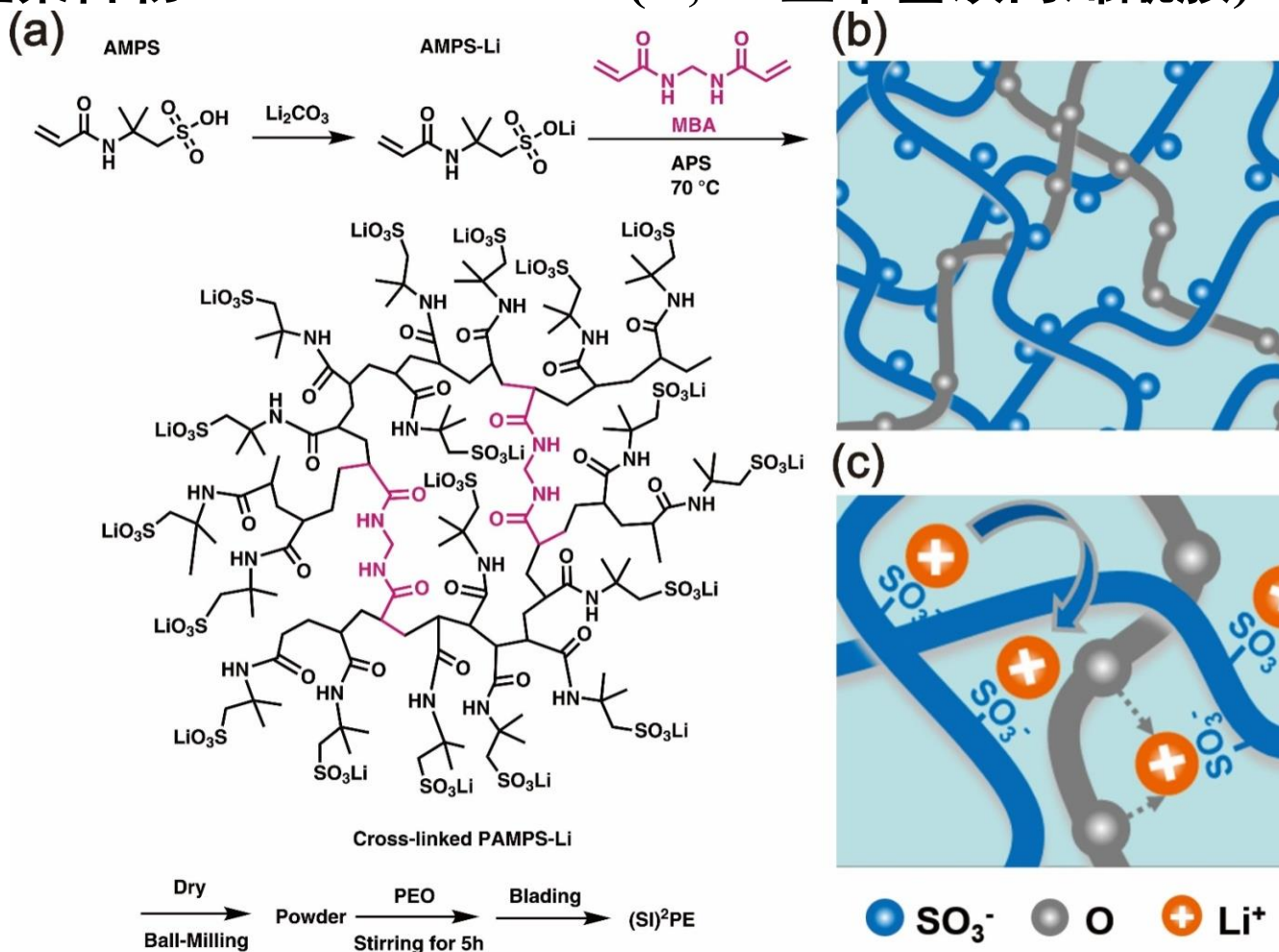
Backbone (PVDF)



Electron withdrawing group
(HFMA)

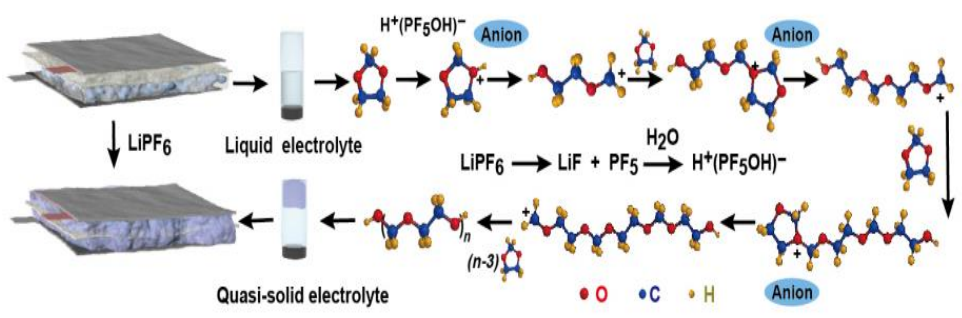
三维半互穿网络结构 (Semi-IPN) 准固态复合单离子导体电解质

交联型锂盐聚合物:AMPS-Li & MBA(N,N'-亚甲基双丙烯酰胺)

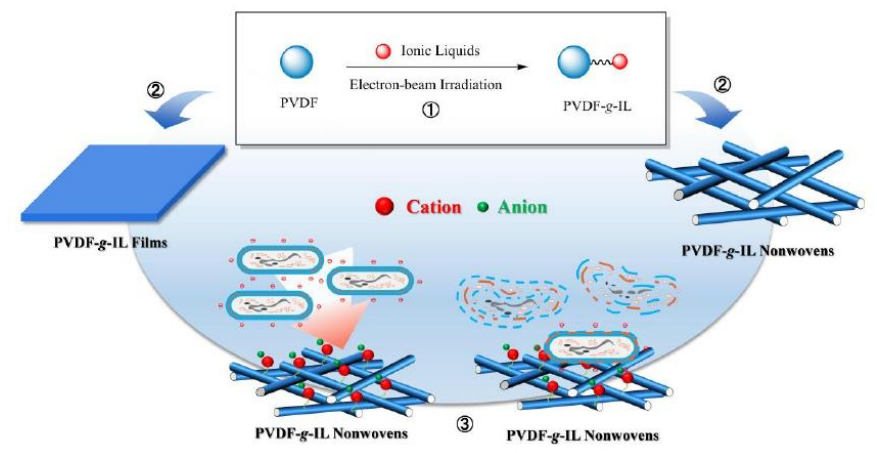


➤ 一步法固化的原位固态化技术

聚合单体+溶剂+电解液原位聚合

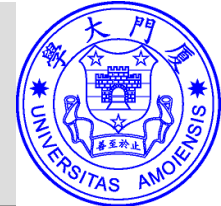


PVDF原位接枝离子液体

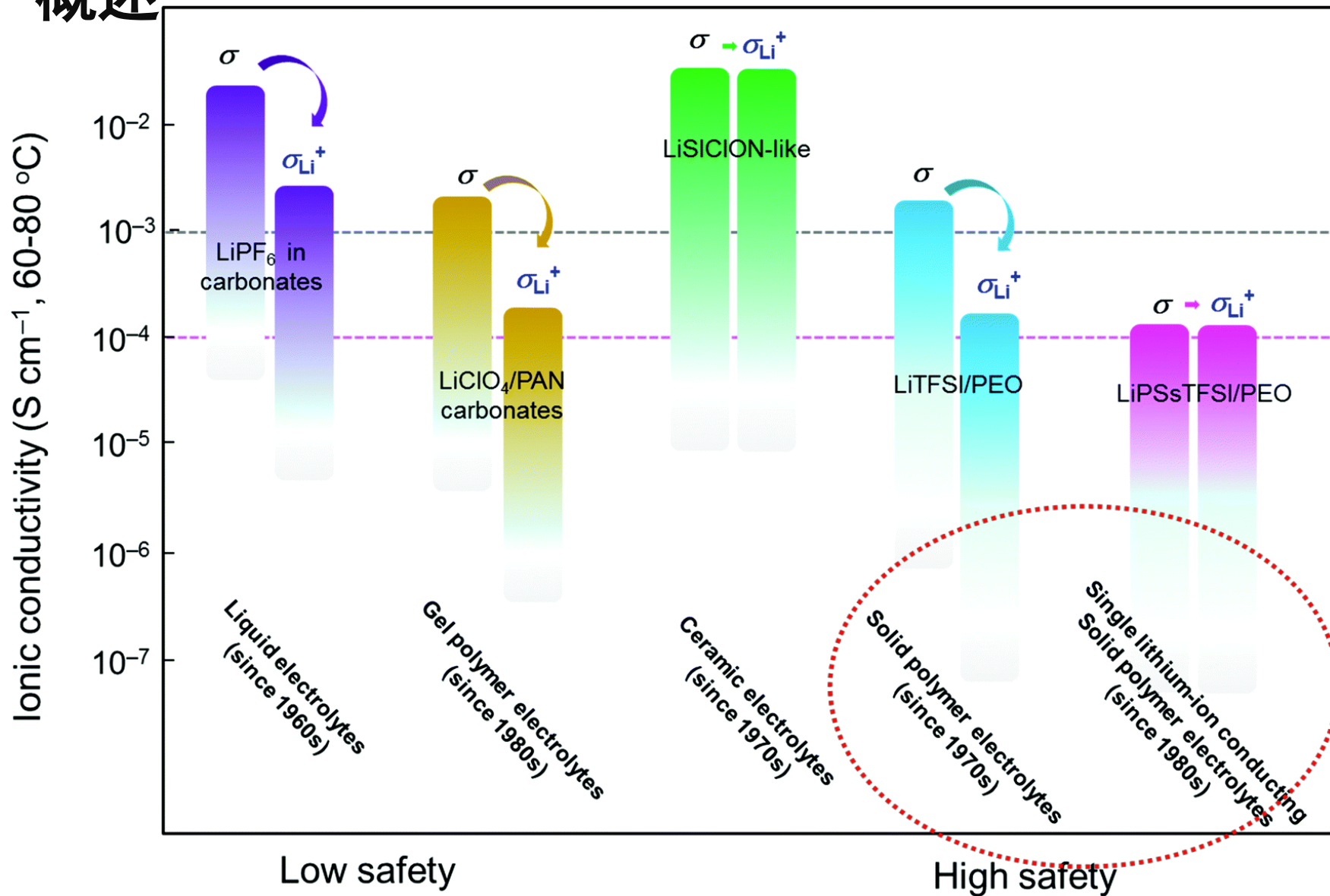


产线与与液态锂电池兼容，后工序阶段增加原位聚合工序，
投资改造成本低，能快速实现大规模生产！

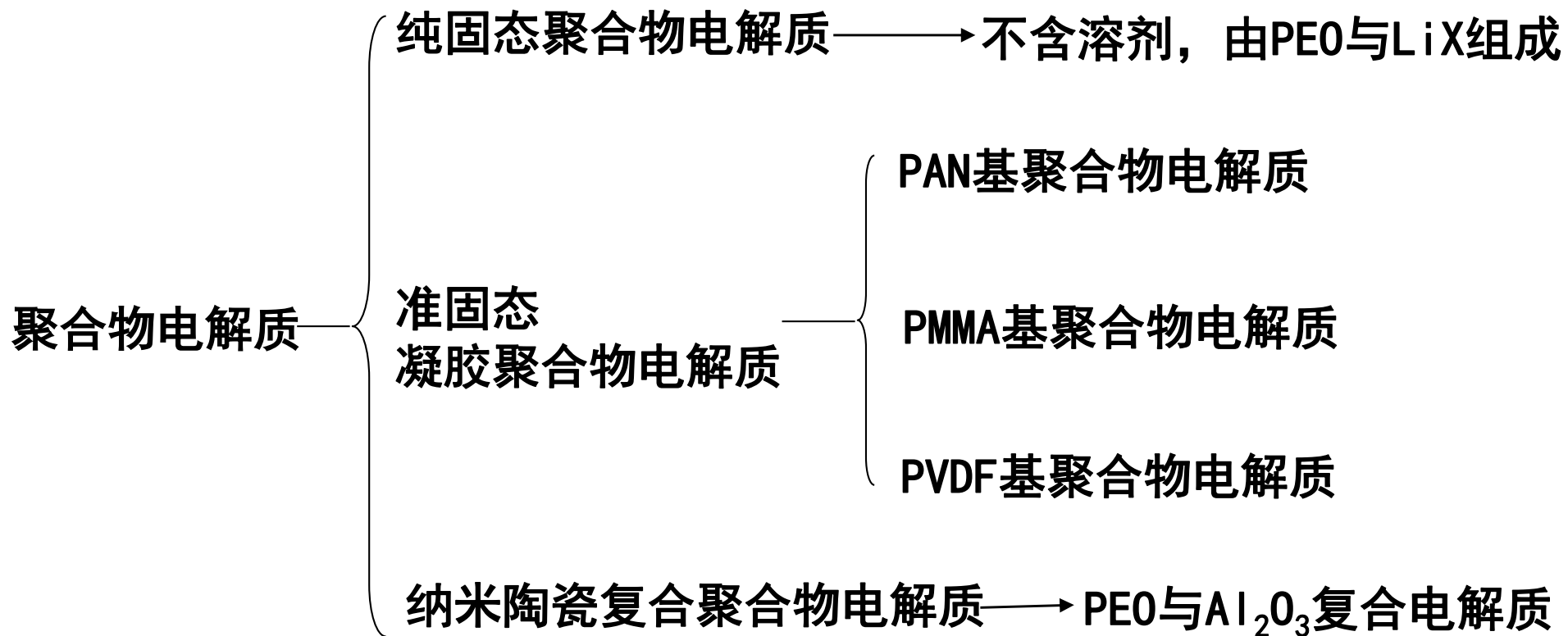
45 全固态聚合物电解质



概述



46 聚合物电解质



中考